

ARTICLE

Gut microbiota-derived TMAVA is a modulator of acute CNS-GVHD

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Acute graft-versus-host disease (aGVHD) can affect the central nervous system (CNS) through microglial activation and T cell infiltration, but the role of gut microbiota in CNS-aGVHD remains unclear. Here, we investigated the role of microbiota in microglial activation during aGVHD using antibiotic-treated specific pathogen-free (SPF), germ-free (GF), and wildling mice. Antibiotic-mediated microbiota depletion led to infiltration of IFN- γ -producing T cells in the brain, activation of microglia via the TLR4/p38 MAPK pathway, and neurocognitive deficits in SPF aGVHD mice. Microglial depletion reversed the neurocognitive deficits. GF and wildling mice treated with antibiotics exhibited similar microglial activation after allogeneic hematopoietic cell transplantation (allo-HCT). Mechanistically, the bacteria-derived metabolite N,N,N-trimethyl-5-aminovaleric acid (TMAVA) was decreased in microglia following antibiotic treatment. TMAVA administration suppressed TLR4/p38 MAPK pathway activity in microglia and alleviated gut microbiota depletion-mediated neurocognitive deficits. Additionally, TMAVA abundance decreased in patient blood after allo-HCT and after GVHD onset. In summary, we identify TMAVA loss as a central causative factor for CNS-aGVHD, opening new perspectives for a metabolite-based therapy.

Introduction

Allogeneic hematopoietic cell transplantation (allo-HCT) is a leading treatment modality for hematological diseases (Giralt and Bishop, 2009). However, the efficacy of allo-HCT is limited by the development of acute graft-versus-host disease (aGVHD), which arises in 30–50% of patients undergoing allo-HCT (Zeiser and Blazar, 2017). aGVHD mainly results from donor-derived T cells, which recognize antigens presented on host antigen-presenting cells (APCs) and cause subsequent

tissue damage by inducing T cell-mediated apoptosis (Blazar et al., 2012; Zeiser and Blazar, 2017). Clinical and preclinical studies have shown that in addition to classical target organs, the central nervous system (CNS) is also affected by aGVHD (Hartrampf et al., 2013; Mathew et al., 2020; Vinnakota and Zeiser, 2021). Previous work in murine aGVHD models has demonstrated infiltration of donor-derived cells in the brain following allo-HCT, as well as the essential role of microglia in

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the development of CNS-aGVHD (Adams et al., 2022; Hartrampf et al., 2013; Mathew et al., 2020; Sailor et al., 2022).

Since patients receiving allo-HCT or those developing aGVHD often undergo antibiotic treatment to manage neutropenic fever and bacterial infections (Freifeld et al., 2011; Horowitz et al., 2021; Styczynski et al., 2018), we aimed to investigate whether antibiotics induce neurological deficits and explore the underlying pathomechanisms. Research in both mice and humans has established a protective role of the commensal gut microbiota and microbial metabolites in aGVHD (Mathewson et al., 2016; Tanaka et al., 2020). With the availability of gnotobiotic mouse strains, sophisticated murine isolators, and cutting-edge genome sequencing techniques, it has been shown that the lack of gut microbiota diversity enhances the risk of aGVHD (Holler et al., 2014; Koyama et al., 2023; Liu et al., 2017; Shono et al., 2015; Shono et al., 2016). Moreover, the expansion and loss of certain bacterial species can influence GVHD severity (Koyama et al., 2023; Simms-Waldrip et al., 2017; Stein-Thoeringer et al., 2019; Wu et al., 2020). These effects are mediated by various mechanisms, including luminal hypoxic shifts that lead to the expansion of facultative anaerobes (Seike et al., 2023), microbial bile acid depletion that causes increased farnesoid X receptor signaling (Lindner et al., 2024), and bacteria-derived butyrate loss that impairs intestinal epithelial regeneration (Mathewson et al., 2016; Tanaka et al., 2020). Tissue regenerative approaches using glucagon-like peptide 2 (Norona et al., 2020), defensins (Ruckert et al., 2022), or lipocalin-2 (Czech et al., 2024) have shown promise in mitigating aGVHD and intestinal dysbiosis.

Microglia are yolk sac-derived resident mononuclear macrophages of the brain (Ginhoux et al., 2010; Schulz et al., 2012). They play multifaceted roles in pathogen clearance, synaptic pruning, neuronal regeneration, and dendritic spine plasticity (Colonna and Butovsky, 2017). Previous studies have revealed that the microbiota plays a crucial role in microglial development and that microbial metabolites impact various CNS pathologies, including Alzheimer's disease and aging (Erny et al., 2021; Erny et al., 2015; Mezo et al., 2020; Mossad et al., 2022).

It is not yet clear how gut microbiota modifications induced by antibiotic treatment impact microglial function and CNS-aGVHD. Using a major histocompatibility complex-mismatched murine aGVHD model, we discovered that antibiotic-mediated microbiota depletion in specific pathogen-free (SPF) mice led to increased T cell infiltration in the brain, elevated TLR4/p38 MAPK pathway activation in microglia, and worsened cognitive deficits, which were rescued by pharmacological microglial depletion. We confirmed an activated microglial phenotype in wildling mice and germ-free (GF) mice undergoing allo-HCT, demonstrating a conserved effect of alteration, depletion, or absence of the gut microbiota on microglia in mice developing aGVHD.

N,N,N-trimethyl-5-aminovaleric acid (TMAVA) is a bacteria-derived metabolite that cannot be synthesized by mammalian cells. TMAVA has been shown to regulate physiological processes such as fetal brain development, as well as heart and liver functions by modulating fatty acid oxidation (FAO) and the production of reactive oxygen species (D'Onofrio et al., 2020; Liu et al., 2021; Zhao et al., 2022). We found that TMAVA levels were

lower in the blood of patients developing aGVHD compared with those without aGVHD and in patient blood after allo-HCT and antibiotic treatment compared with levels before allo-HCT. In our murine aGVHD model, TMAVA administration reversed microglial TLR4/p38 MAPK signaling and neurocognitive deficits that were induced by antibiotic-mediated gut microbiota depletion.

Our findings demonstrate that gut microbiota depletion exacerbates CNS-aGVHD in mice in a microglia-dependent manner. Supplementation with the metabolite TMAVA under gut microbiota-depleted conditions reduced the severity of CNS-aGVHD.

Results

Antibiotic-treated aGVHD mice exhibit increased T cell infiltration and activation in the brain

To assess the impact of gut microbiota on the CNS and on classical target organs during aGVHD, an antibiotic combination of vancomycin, gentamicin, cefoxitin, and metronidazole as described previously (Erny et al., 2015) was orally administered to SPF BALB/c mice daily to deplete the gut microbiota. After 14 days of antibiotic treatment (D-14 to D-1), mice were lethally irradiated and underwent allo-HCT on day 0 (D0). The antibiotic treatment was continued for 13 days following allo-HCT (D+1 to D+13) to sustain microbiota depletion (Fig. 1 a). On D+14, mice exhibited enlarged cecum (Fig. 1, b and c), indicative of a depleted gut microbiota. Additionally, on D-1 (after 14 days of antibiotic treatment) (Fig. 1, d-f) and on D+13 (after 27 days of antibiotic treatment) (Fig. 1, g-i), the stool was analyzed for abundance of live bacteria using the SYTO 9 nuclear stain for gram-positive and gram-negative bacteria. We observed a decline in the frequency and absolute numbers of live bacteria at both time points. Additionally, we found reduced levels of commensal gut microbiota-derived short-chain fatty acids (SCFAs) acetate, butyrate, and propionate (Flint et al., 2015) in the colon on D+14 (Fig. 1, j-l). Antibiotic treatment did not change body weight (Fig. 1 m), H2-k^b donor cell engraftment in the bone marrow (Fig. 1 n), or aGVHD severity in the liver and small intestine, but it worsened aGVHD in the colon (Fig. 1, o-q). Furthermore, it led to systemic activation of myeloid cells as the surface expression of CD40 and CD80 was increased on splenic macrophages in antibiotic-treated mice (Fig. 1, r and s). This was consistent with other studies that demonstrated increased aGVHD severity in GF mice compared with SPF mice (Seike et al., 2023).

Donor-derived immune cell infiltration into tissues is a hallmark of aGVHD (Zeiser and Blazar, 2017). Therefore, we sought to investigate whether antibiotics alter immune cell infiltration into the brain after allo-HCT. Antibiotic-treated aGVHD mice exhibited higher percentages and numbers of donor-derived CD3⁺ CD45⁺ T cells in the brain (Fig. 2, a-c; and Fig. S1, a and b). We confirmed increased numbers of CD3⁺ cells in the cortex and cortical meninges by immunohistochemistry (Fig. 2, d-f), but not in the cerebellum or hippocampus (Fig. S1, c and d). The number of T cells transplanted into mice on D0 correlated with the frequency of T cells infiltrating the brain

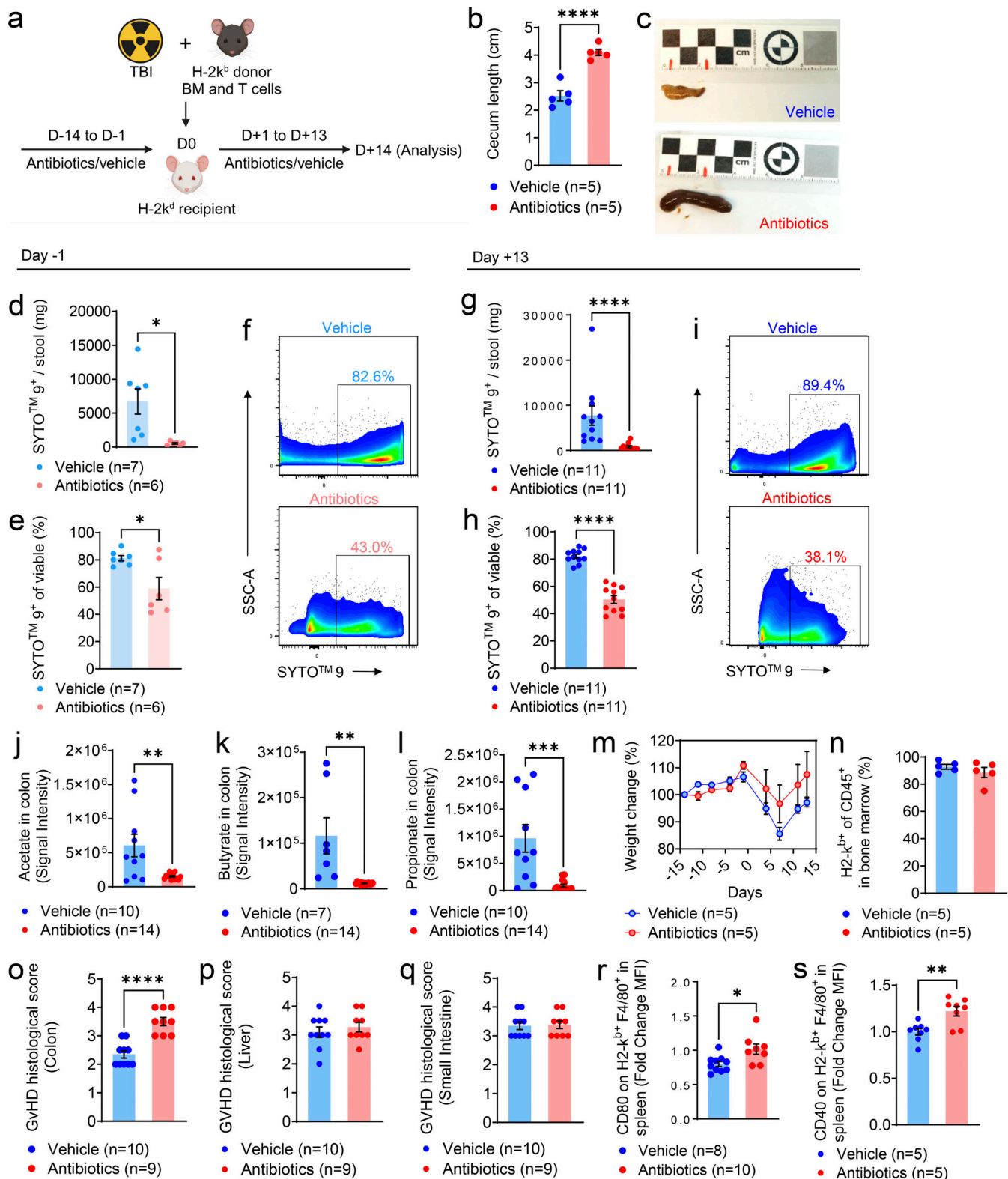


Figure 1. Antibiotic-mediated microbiota depletion in aGVHD exacerbates systemic inflammation. (a–n) Analysis of cecum, colon, stool, and bone marrow cells on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). **(a)** Schematic depiction of treatment and allo-HCT. **(b–l)** Quantification of cecum length **(b)** and **(c)** representative images of each group. Quantification of **(d)** SYTO 9⁺ bacteria normalized to stool weight and **(e)** percentage of SYTO 9⁺ bacteria using flow cytometry, along with **(f)** representative flow cytometry plots of each group on D-1 (after 14 days of antibiotics/vehicle treatment). Quantification of **(g)** SYTO 9⁺ bacteria normalized to stool weight and **(h)** percentage of SYTO 9⁺ bacteria using flow cytometry, along with **(i)** representative flow cytometry plots of each group on D+13 (after 27 days of antibiotics/vehicle treatment). Quantification of **(j)** acetate, **(k)** butyrate, and **(l)** propionate in colon using targeted mass spectrometry analysis of polar metabolites. **(m)** Quantification of percentage of body weight of mice

at different time points compared with the weight at start of antibiotic treatment (D–14). **(n)** Quantification of percentage of H2-k^{b+} cells in the bone marrow using flow cytometry. The experiment was performed once (b, c, m, and n) or two times (d–l) and results (mean ± SEM) pooled. *n* = 5–11 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test, the Mann-Whitney *U* test, or two-way ANOVA with Sidák's multiple comparisons test. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001. **(o–q)** Histological scoring of GVHD severity in colon, liver, and small intestine samples stained for H&E on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of GVHD histological scores in (o) colon, (p) liver, and (q) small intestine. Experiments were performed two times and results (mean ± SEM) pooled. *n* = 9 or 10 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. *****P* < 0.0001. **(r and s)** Analysis of the spleen on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of (r) CD80 and (s) CD40 on H2-k^{b+} F4/80⁺ cells in the spleen using flow cytometry. The experiment was performed once (s) or two times (r) and results (mean ± SEM) pooled. *n* = 5–10 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test or the Mann-Whitney *U* test. **P* < 0.05, ***P* < 0.01.

(Fig. 2, g and h), indicating that gut microbiota depletion and abundance of transplanted T cells both contribute to T cell infiltration into the brain. We next analyzed the phenotype of brain-infiltrating T cells and observed an increase in the frequency of central memory T cell population (CD44⁺ CD62L⁺) (Fig. 2, i and j; and Fig. S1 e). The ratio of CD8⁺ T cells to CD4⁺ T cells was also increased (Fig. 2 k), in line with a reduction in the frequency of CD4⁺ T cells (Fig. S1, f–i). Analysis of T cell functional markers revealed that both the CD8⁺ and CD4⁺ T cell compartments in the brains of antibiotic-treated mice expressed higher intracellular levels of the pro-inflammatory cytokine IFN-γ (Fig. 2, l–o). Numbers of CD4⁺ IFN-γ⁺ and CD8⁺ IFN-γ⁺ cells were also increased in the brains of mice treated with antibiotics compared with vehicle during aGVHD (Fig. S1, j and k). The surface expression of activation markers CD69 and CD25 was increased on T cells in the antibiotics group (Fig. 2, p–s). To assess whether gut microbiota depletion impacts the CNS irrespective of occurrence of aGVHD, we analyzed mice treated with antibiotics as described previously, which underwent either syngeneic HCT (syn-HCT) or allo-HCT. Allo-HCT mice treated with antibiotics exhibited higher frequencies of T cells in the brain as compared to their syn-HCT counterparts (Fig. S1 l). This demonstrated that increased T cell infiltration in the brains of antibiotic-treated allo-HCT mice was not solely an antibiotic-mediated effect.

Apart from donor-derived T cells, neutrophils, monocytes, dendritic cells (DCs), and B cells also modulate aGVHD severity by affecting T cell alloreactivity, cytokine production, and integrin-mediated cell homing (Hülsdunker et al., 2018; Jardine et al., 2020; Jeon et al., 2022; Koyama et al., 2015). Therefore, we analyzed different myeloid and lymphoid cell subsets in the brain. The frequency of CD11b⁺ CD45^{hi} myeloid cells, monocytes, DCs, and B cells was similar in the brain when comparing antibiotic- and vehicle-treated groups (Fig. S1, m–p). We observed an increase in the frequency of CD11b⁺ Ly6G⁺ neutrophils (Fig. 2, t and u), and CD206⁺ CNS-associated macrophages (CAMs) and perivascular macrophages (Fig. 2, v and w) in the brains of antibiotic-treated mice. Absolute numbers of neutrophils and CD206⁺ cells were similar in both groups (Fig. S1, q and r).

Previous findings have demonstrated that whole-brain radiation disrupts the blood-brain barrier (BBB) (Blethen et al., 2023; Zhang et al., 2024). Therefore, we investigated whether increased T cell infiltration in the brains of antibiotic-treated mice was mediated by higher endothelial extravasation or compromised tight junction integrity. We utilized CD31 as a

marker for endothelial cells and found no differences in percentages of CD31⁺ CD45⁺ cells when comparing antibiotic-treated versus vehicle-treated groups (Fig. S1 s). The expression of endothelial E-cadherin, which is important for the maintenance of tight junction and BBB integrity (Kiptoo et al., 2011), was decreased in antibiotic-treated mice (Fig. 2, x and y). The expression of other adhesion markers like VCAM-1, ICAM-1, ICAM-2, occludin-1, and claudin-1 was not altered in antibiotic-treated versus vehicle-treated mice (Fig. S1, t–x). To assess whether increased T cell infiltration in the brain is due to aGVHD-mediated systemic inflammation, we analyzed T cells in the blood and spleen. CD4⁺ and CD8⁺ T cells in the blood of antibiotic-treated mice had increased surface expression of CD69 (Fig. S2, a and b). Apart from that, we found no differences in either frequencies of T cells or IFN-γ production by T cells in the blood or spleen of antibiotic- versus vehicle-treated groups (Fig. S2, c–l). Brain weights between the groups remain unchanged, indicating a lack of atrophy or edema (Fig. S2 m).

Antibiotic-treated aGVHD mice exhibit increased microglial proliferation and activation

Previous research has shown that microglial development and functions in health and disease are affected when the gut microbiota is altered (Erny et al., 2021; Erny et al., 2015). Therefore, we sought to delineate the impact of microbiota depletion on microglia in mice developing aGVHD. SPF BALB/c mice were treated with antibiotics and underwent allo-HCT as described previously (Fig. 1 a). We observed higher frequencies and numbers of CD11b⁺ CD45^{int} myeloid cells in the brain of antibiotic-treated compared with vehicle-treated mice on D+14 (Fig. 3, a–c). CD11b⁺ CD45^{int} cells expressed the microglia-specific marker P2RY12 and were recipient-derived as evidenced by the expression of H2-k^d (Fig. S2, n–p), confirming that this population was tissue-resident microglia. We did not observe a difference in percentages of CD11b⁺ CD45^{int} cells in the brains of antibiotic-treated mice undergoing allo-HCT compared with syn-HCT (Fig. S2 q). Using Iba-1 as a myeloid cell marker, we observed increased numbers of Iba-1⁺ cells in the cerebral cortex of antibiotic-treated versus vehicle-treated allo-HCT mice, whereas numbers of Iba-1⁺ cells in the cerebellum and hippocampus were similar in the two groups (Fig. 3, d and e; and Fig. S2, r and s). To assess whether increased abundance of microglia in the brain is connected to increased cell proliferation, we analyzed Ki67 expression of Iba-1⁺ cells using immunofluorescence. We found an increase in numbers of Iba-1⁺ Ki67⁺

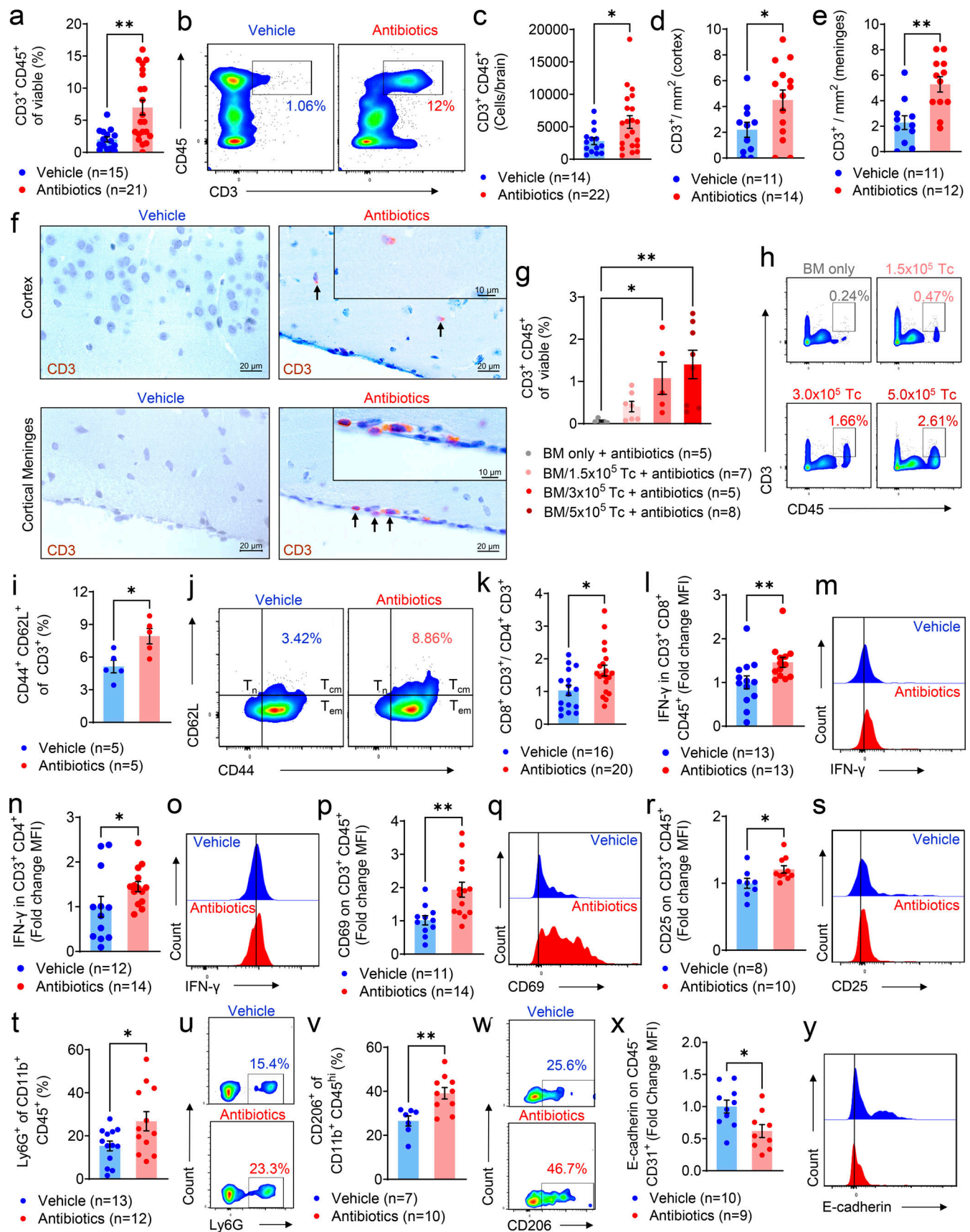


Figure 2. **Antibiotics increase immune cell infiltration in the brain during aGVHD.** (a–c) Analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^d) recipients treated with antibiotics or vehicle (water). (a and b) Quantification of the percentage of CD3⁺ CD45⁺ cells using flow cytometry (a) along

with (b) representative flow cytometry plots of each group. (c) Quantification of absolute numbers of CD3⁺ CD45⁺ cells. Experiments were performed four (a and b) or five times (c) and results (mean $\pm$ SEM) pooled. $n = 14$ –22 per group. Each data point represents a biological replicate. P values were calculated using the Mann–Whitney *U* test. * $P < 0.05$, ** $P < 0.01$. (d–f) Histology of brain samples stained for CD3⁺ cells on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). The scatter plot shows absolute number (per mm²) of CD3⁺ cells in the (d) cortex and (e) cortical meninges, along with (f) representative images of each group (scale bars, 20 μ m; inset, 10 μ m). Experiments were performed three times and results (mean $\pm$ SEM) pooled. $n = 11$ –14 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test or the Mann–Whitney *U* test. * $P < 0.05$, ** $P < 0.01$. (g and h) Analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics and transplanted on D0 with different T cell numbers. (g) Quantification of the percentage of CD3⁺ CD45⁺ cells using flow cytometry along with (h) representative flow cytometry plots of each group. Experiments were performed two times and results (mean $\pm$ SEM) pooled. $n = 11$ –14 per group. Each data point represents a biological replicate ($n = 1$). P values were calculated using the Kruskal–Wallis test. * $P < 0.05$, ** $P < 0.01$. (i–y) Flow cytometry analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). (i) Quantification of the percentage of CD44⁺ CD62L⁺ cells along with (j) representative flow cytometry plots of indicated groups. Quantification of the (k) ratio of CD3⁺ CD8⁺ cells to CD3⁺ CD4⁺ cells, (l) IFN- γ in CD3⁺ CD8⁺ cells, (n) IFN- γ in CD3⁺ CD4⁺ cells, (p) CD69 on CD3⁺ CD45⁺ cells, (r) CD25 on CD3⁺ CD45⁺ cells, (t) percentage of Ly6G⁺ cells, and (v) CD206⁺ cells, and (x) E-cadherin on CD31⁺ CD45⁺ cells. Representative histograms of (m) IFN- γ in CD3⁺ CD8⁺ cells, (o) IFN- γ in CD3⁺ CD4⁺ cells, (q) CD69 on CD3⁺ CD45⁺ cells, and (s) CD25 on CD3⁺ CD45⁺ cells and representative flow cytometry plots of (u) Ly6G⁺ and (w) CD206⁺ cells in each group. The experiment was performed once (i and j), two times (r, s, v, w, x, and y), three times (p, q, t, and u), or four times (k–o) and results (mean $\pm$ SEM) pooled. $n = 5$ –20 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test or the Mann–Whitney *U* test. * $P < 0.05$, ** $P < 0.01$.

cells in the brains of antibiotic-treated mice (Fig. 3, f and g). Microarray analysis of sorted CD11b⁺ CD45^{int} cells confirmed enrichment of gene sets related to cell proliferation pathways in antibiotic-treated mice (Fig. 3 h). We further assessed the phenotype of CD11b⁺ CD45^{int} cells by analyzing them for the surface expression of CD11c, a marker for activation, and P2RY12, a marker for homeostasis. We found increased CD11c expression and decreased P2RY12 expression in the antibiotic-treated group (Fig. 3, i–l). Together, this suggests that antibiotic treatment resulted in an activated and proliferative microglial phenotype.

TLR4-mediated brain inflammation plays a decisive role in many CNS pathologies by modulating myelination, BBB integrity, and phagocytosis (Choi et al., 2020; Wu et al., 2022). Here, we observed that CD11b⁺ CD45^{int} and CD11b⁺ CD45^{hi} myeloid cells had elevated surface expression of TLR4 in antibiotic-treated compared with vehicle-treated allo-HCT mice (Fig. 3, m and n; and Fig. S2 t). Further, intracellular levels of MyD88 were higher in mice transplanted with a higher number of T cells on D0 (Fig. 3, o and p). We also observed a positive correlation between TLR4 expression on CD11b⁺ CD45^{int} cells and T cell percentages in the brains of antibiotic-treated mice (Fig. 3 q). When we analyzed the expression of costimulatory molecules on CD11b⁺ CD45^{int} cells, we found no changes in the expression of MHC II, CD80, and CD40 between antibiotics and vehicle-treated allo-HCT mice (Fig. S2, u–w). Antibiotic-treated mice undergoing allo-HCT exhibited the decreased surface expression of P2RY12, but showed no changes in TLR4 and CD11c expression on CD11b⁺ CD45^{int} cells compared with syn-HCT mice (Fig. S2, x–z).

Since antibiotic treatment was connected to more severe aGVHD in the colon, we assessed the colon lamina propria and epithelium for changes in infiltrating immune cells and their TLR4 expression. We found no changes in frequencies of different myeloid and lymphoid cell subsets, or in the expression of TLR4 on tissue-resident macrophages between the antibiotic and vehicle-treated groups in either colon compartment (Fig. S3, a–l). Next, we evaluated whether increased microglial TLR4 expression could result from increased infiltration of TLR4 ligands into the colon. Lipopolysaccharide (LPS) is one of the best

characterized TLR4 ligands (Kim et al., 2023). Using an antibody for the lipid A moiety of LPS, we quantified abundance of LPS in the colon tissue by immunofluorescence. The absolute number of lipid A⁺ foci in the colon tissue was comparable between the two groups (Fig. S3, m and n).

To understand whether the changes in microglial phenotype after antibiotic treatment are accompanied by changes in gene transcription, we performed microarray analysis of sorted CD11b⁺ CD45^{int} cells and observed an upregulation of inflammation-related genes like *Nfkb1*, *Irgax*, *Traf2*, *Traf3*, *Nod2*, *Trim12c*, *Irf7*, and *Irak2* in antibiotic- versus vehicle-treated allo-HCT mice (Fig. 4 a). IFN- γ response, MAPK pathway, and NF- κ B pathway gene sets were also upregulated (Fig. 4, b–d), and we confirmed an increase in intracellular phospho-p38 MAPK and phospho-NF- κ B p65 (Fig. 4, e and f) using flow cytometry. To understand the role of TLR4 in microglial activation and intracellular phospho-kinase levels, SPF BALB/c mice were lethally irradiated and underwent allo-HCT and antibiotics regimen as described previously. In addition, the mice were treated with a TLR4 inhibitor (CLI-095) or vehicle (3% vol/vol DMSO) via daily intraperitoneal injection (D+1 to D+13) (Fig. 4 g). Commensurate with previous studies in TLR4-deficient mice (Liang et al., 2014; Zhao et al., 2013), treatment with CLI-095 reduced GVHD severity in the colon, liver, and small intestine (Fig. 4, h–j) without affecting H2-k^{b+} donor cell engraftment in the bone marrow (Fig. 4 k). CLI-095-treated mice also showed a reduction in intracellular levels of MyD88 and phospho-p38 MAPK in CD11b⁺ CD45^{int} cells (Fig. 4, l and m), indicating that abundance of phospho-p38 MAPK and MyD88 is TLR4 dependent. The surface expression of CD69 was diminished on brain-infiltrating CD3⁺ cells (Fig. 4 n), but the frequency of infiltrating CD3⁺ CD45⁺ T cells and their intracellular IFN- γ levels remained unchanged upon CLI-095 treatment (Fig. 4, o and p). To assess systemic changes caused by CLI-095 treatment, we also analyzed T cells in the blood. We did not observe changes in frequencies of T cells or CD69 expression on T cells between the two groups (Fig. 4, q and r), but intracellular IFN- γ levels in circulating CD8⁺ T cells were reduced (Fig. 4 s). Taken together, these findings support the concept that pharmacological TLR4

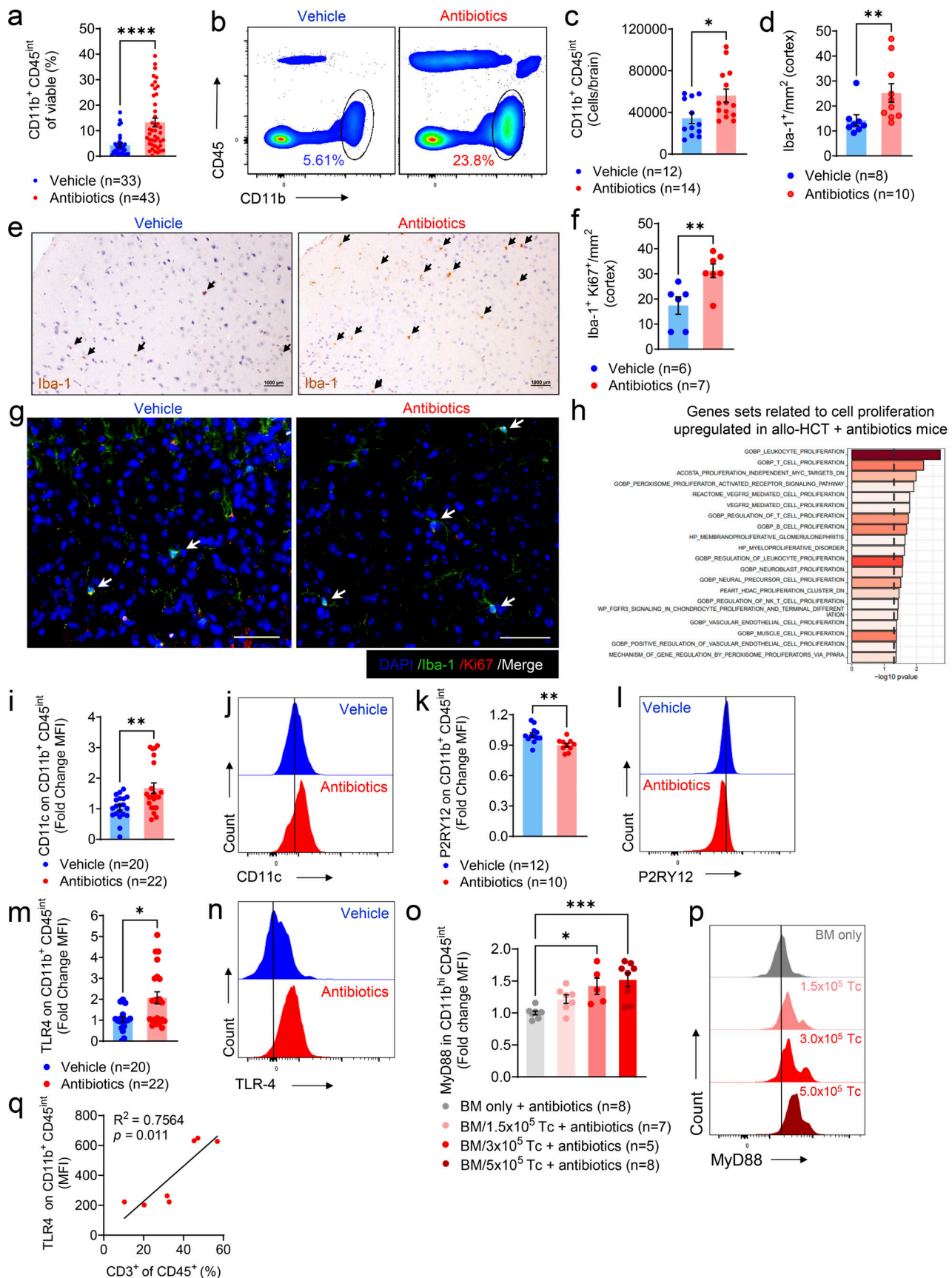


Figure 3. **Microbiota depletion increases microglial activation in aGVHD.** (a–c) Flow cytometry analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). (a and b) Quantification of the percentage of CD11b⁺ CD45^{int} cells (a) along with (b)

representative flow cytometry plots of each group. **(c)** Quantification of absolute numbers of CD11b⁺ CD45^{int} cells. Experiments were performed three (c) or nine times (a and b) and results (mean $\pm$ SEM) pooled. $n = 12$ –43 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test or the Mann–Whitney *U* test. * $P < 0.05$, **** $P < 0.0001$. **(d–g)** Histology of brain samples stained for Iba-1⁺ cells and Iba-1⁺ Ki67⁺ cells analyzed on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of the absolute number (per mm²) of (d) Iba-1⁺ and (f) Iba-1⁺ Ki67⁺ cells in the cerebral cortex, along with representative images of (e) Iba-1⁺ cells (scale bars, 1,000 μ m) and (g) Iba-1⁺ Ki67⁺ cells (scale bars, 50 μ m) in each group. Experiments were performed two (f and g) or three times (d and e) and results (mean $\pm$ SEM) pooled. $n = 8$ –10 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. ** $P < 0.01$. **(h)** Gene set enrichment analysis on microarray analysis shows differentially regulated cell proliferation-related pathways from FACS-sorted CD11b⁺ CD45^{int} cells isolated from the brains of SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics ($n = 2$) or vehicle ($n = 2$) and analyzed on D+14 after allo-HCT. The experiment was performed once. **(i–n)** Flow cytometry analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of (i) CD11c, (k) P2RY12, and (m) TLR4 on CD11b⁺ CD45^{int} cells along with representative histograms of (j) CD11c, (l) P2RY12, and (n) TLR4 in each group. Experiments were performed three (k and l) or six times (i, j, m, and n) and results (mean $\pm$ SEM) pooled. $n = 10$ –22 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test or the Mann–Whitney *U* test. * $P < 0.05$, ** $P < 0.01$. **(o–p)** Flow cytometry analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics and transplanted on D0 with different T cell numbers. **(o and p)** Quantification of MyD88 in CD11b⁺ CD45^{int} cells (o) along with (p) representative histograms of each group. Experiments were performed two times and results (mean $\pm$ SEM) pooled. $n = 5$ –8 per group. Each data point represents a biological replicate. P values were calculated using ordinary one-way ANOVA with Tukey's multiple comparisons test. * $P < 0.05$, *** $P < 0.001$. **(q)** Flow cytometry analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics. Correlation analysis using simple linear regression between TLR4 (MFI) on CD11b⁺ CD45^{int} cells and percentage of CD3⁺ CD45⁺ cells in the brain. Experiments were performed two times and results pooled.

inhibition modulates CNS inflammation concurrent with a reduction of aGVHD severity in classical target organs.

Antibiotic-treated aGVHD mice develop cognitive deficits

Microglia produce and release a diverse range of molecules including cytokines and neurotransmitters that can directly impact neuronal activity and cognitive functions (Bechade et al., 2013). To investigate alterations in neurotransmission-related genes in microglia after antibiotic treatment, we performed a microarray analysis of sorted CD11b⁺ CD45^{int} cells from brains of SPF BALB/c mice that had undergone allo-HCT and antibiotics or vehicle treatment as described previously (Fig. 1 a). We observed a downregulation of genes related to postsynaptic neurotransmission in the antibiotic-treated group, notably genes encoding GABA and glutamate receptors like *Gabra6*, *Gabra3*, *Gabra2*, *Grin3a*, *Grik1*, *Grik2*, *Grik3*, *Grik4*, and *Grik5* (Fig. 5 a). Using targeted mass spectrometry analysis for polar metabolites, we confirmed a decline in intracellular levels of excitatory neurotransmitters aspartate and glutamate, as well as inhibitory neurotransmitter GABA in sorted CD11b⁺ CD45^{int} cells (Fig. 5, b–d). Qualitative analysis of brain tissues stained with Luxol fast blue/periodic acid–Schiff (LFB–PAS) did not show differences in myelination in the cortex, cerebellum, or hippocampus between the groups (Fig. S4 a), indicating that cognitive deficits observed in antibiotic-treated mice are not due to changes in myelination in different brain regions.

Next, we assessed whether cognitive parameters are affected in antibiotic- versus vehicle-treated aGVHD mice and we functionally validated the role of microglia in the development of cognitive deficits induced by antibiotic treatment. The maintenance of adult microglial populations requires constitutive colony-stimulating factor-1 receptor (CSF1R) signaling (Elmore et al., 2014). SPF C57BL/6 mice were maintained (D–21 to D+19) either on a brain-penetrant CSF1R inhibitor (PLX5622)–formulated chow to deplete microglia or on a control chow (Spangenberg et al., 2019). Additionally, these mice underwent allo-HCT. The microglia-depleted mice were treated with antibiotics, and the nondepleted mice were treated with either

antibiotics or vehicle ad libitum via drinking water (Fig. 5 e). For the cognitive tests, we used C57BL/6 recipients as these mice exhibit lower anxiety, greater resilience to stress, and enhanced emotional stability compared with BALB/c mice (Heinla et al., 2018; Razzoli et al., 2011). On D+19, a decrease in percentages and numbers of CD11b⁺ CD45^{int} myeloid cells in the brains of mice maintained on PLX5622 chow versus control chow confirmed microglial depletion (Fig. 5, f–h). Behavior tests on D+14 to D+19 revealed a decrease in alternations in the T-maze test (Fig. 5 i) indicating a decline in short-term memory, a decrease in time spent at the novel object in the novel object recognition test indicating a decline in long-term memory (Fig. 5 j), and a reduction of entries into the open arms of an elevated plus maze test, indicating increased anxiety (Fig. 5 k), in antibiotic- versus vehicle-treated mice. These antibiotic-mediated deficits were reversed upon pharmacological microglial depletion in mice maintained on PLX5622 chow versus control chow (Fig. 5, i–k). Additionally, antibiotic-treated mice maintained on PLX5622 chow had improved spatial memory compared with antibiotic-treated mice maintained on control chow, as well as vehicle-treated mice maintained on control chow, as they performed better in the clock maze test (Fig. 5 l). CSF1R signaling has also been shown to regulate the development and function of other myeloid cell subtypes (Stanley and Chitu, 2014). We found no changes in the frequencies of monocytes, neutrophils, DCs, or T cells in the brain, spleen, and bone marrow of microglia-depleted versus nondepleted groups except for a decrease in the frequency of Ly6C⁺ monocytes in the bone marrow (Fig. S4, b–k). Behavior tests with antibiotic-treated mice undergoing syn-HCT compared with allo-HCT did not reveal any difference in alternations in the T-maze test or in time spent at the novel object in the novel object recognition test, but showed a decrease in entries into the open arms of an elevated plus maze test (Fig. S4, l–n). We also confirmed that the observed cognitive changes were not attributable to impaired vision or muscular weakness caused by aGVHD (Fig. S4, o and p), as no significant differences were detected in time spent in the safe zone during the visual cliff test or in grip strength measurements between the three tested groups.

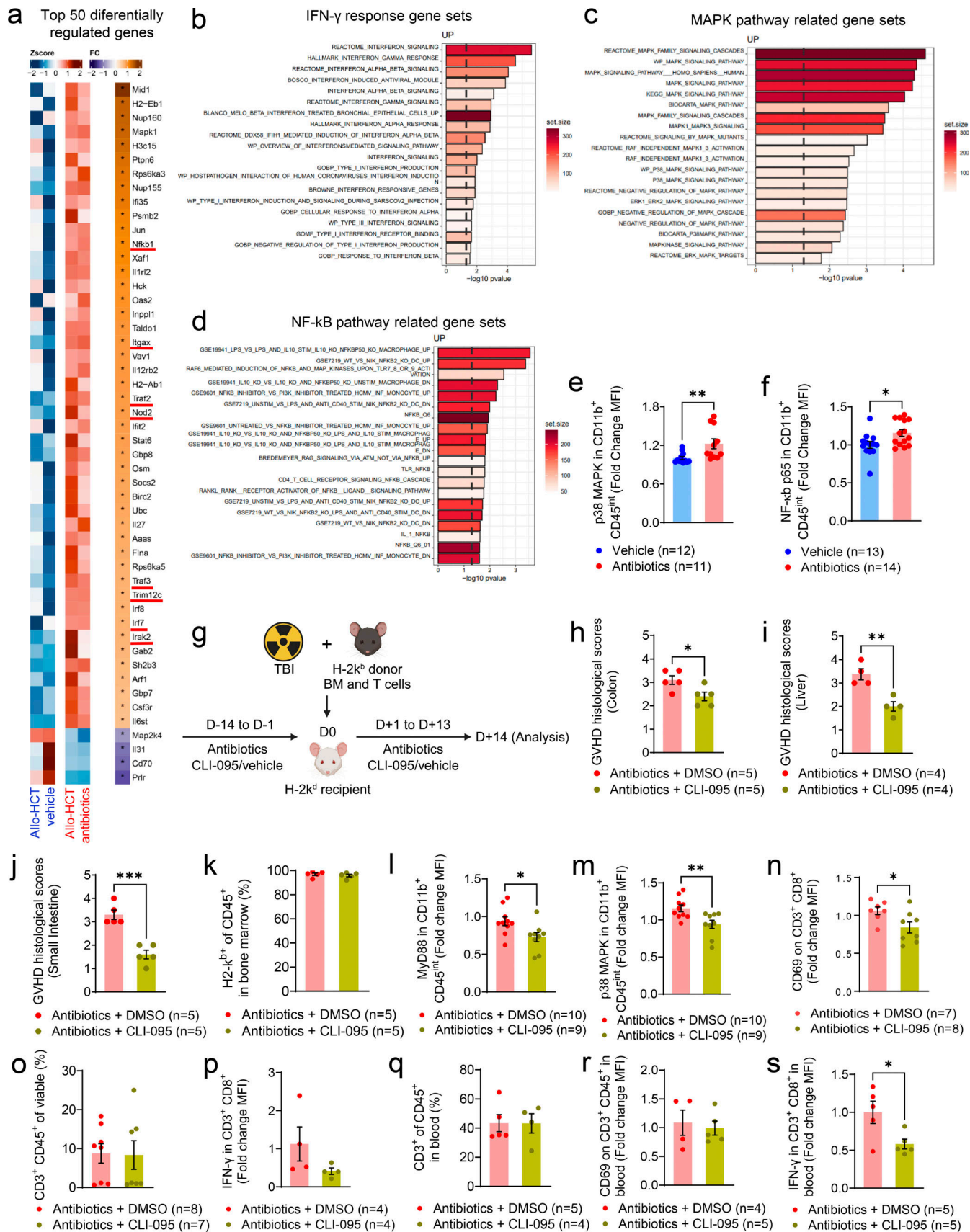


Figure 4. **Microbiota depletion upregulates inflammatory pathways in microglia after allo-HCT.** (a–d) Microarray analysis of FACS-sorted CD11b⁺ CD45^{int} cells from SPF BALB/c (H2-k^d) recipients treated with antibiotics ($n = 2$) or vehicle ($n = 2$), analyzed on D+14 after allo-HCT. (a) Heat map shows the

top 50 differentially regulated genes from "REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SYSTEM" (*). Color code represents the Z-score $\log_2$ intensity (left panel) and $\log_2$ fold change (right panel). Bar plots show upregulated (b) IFN- γ response gene sets, (c) MAPK pathway-related gene sets, and (d) NF- κ B pathway-related gene sets in the antibiotic-treated group. The experiment was performed once. **(e and f)** Flow cytometry analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of (e) phospho-p38 MAPK (Thr180, Tyr182) and (f) phospho-NF- κ B p65 (Ser536) in CD11b⁺ CD45^{int} cells. Experiments were performed three times and results (mean $\pm$ SEM) pooled. $n = 11$ –14 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. **(g–j)** Histological scoring of GVHD severity in colon, small intestine, and liver samples stained for H&E on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics and CLI-095 or vehicle (DMSO). **(g)** Schematic depiction of treatment and allo-HCT. GVHD histological scores in (h) colon, (i) liver, and (j) small intestine. The experiment was performed once. $n = 4$ –5 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. **(k)** Flow cytometry analysis of bone marrow cells on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics (all animals) and CLI-095 or vehicle (DMSO). Quantification of percentage of H2-k^{b+} cells in the bone marrow. Experiments were performed two times and results (mean $\pm$ SEM) pooled. $n = 5$ per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. **(l–p)** Flow cytometry analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics (all animals) and CLI-095 or vehicle (DMSO). Quantification of (l) MyD88 in CD11b⁺ CD45^{int} cells, (m) phospho-p38 MAPK (Thr180, Tyr182) in CD11b⁺ CD45^{int} cells, (n) CD69 on CD3⁺ CD8⁺ cells, (o) percentage of CD3⁺ CD45⁺ cells, and (p) IFN- γ in CD3⁺ CD8⁺ cells. Experiments were performed two times and results (mean $\pm$ SEM) pooled. $n = 4$ –10 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired t test or the Mann-Whitney U test. * $P < 0.05$, ** $P < 0.01$. **(q–s)** Flow cytometry analysis of PBMCs isolated from the blood on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients, treated with antibiotics and CLI-095 (all animals) or vehicle (DMSO). Quantification of (q) percentage of CD3⁺ CD45⁺ cells, (r) CD69 on CD3⁺ CD45⁺ cells, and (s) IFN- γ in CD3⁺ CD8⁺ cells. The experiment was performed once. $n = 4$ –5 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. * $P < 0.05$.

These findings suggest that cognitive deficits in microbiota-depleted aGVHD mice are solely microglia-dependent and can be reversed upon pharmacological microglial depletion via CSF1R inhibition.

Microglial activation is conserved across different gnotobiotic murine models of aGVHD

To validate the reproducibility of our findings in mice with a different, more diverse gut microbiota compared with SPF mice, we used wildling recipients, whose gut microbiota is derived from wild mice and recapitulates the complexity and diversity of the human microbiota (Rosshart et al., 2019). Wildling BALB/c mice underwent lethal irradiation followed by allo-HCT. Additionally, one group was treated with either a single antibiotic, cefepime (D+1 to D+10), more closely resembling clinical practice (Boyd et al., 2019) via daily subcutaneous injection, and another group was untreated. Both groups were analyzed on D+14 (Fig. 6 a). Cefepime treatment reduced the number of live bacteria in the stool on D+14 (Fig. 6 b). Analysis of the brain revealed that CD11b⁺ CD45^{int} myeloid cells of cefepime-treated mice showed the increased surface expression of MHC II, CD11c, and F4/80, as well as higher intracellular levels of phospho-NF- κ B p65 (Fig. 6, c–f). Percentages of CD11b⁺ CD45^{int} myeloid cells and their surface expression of TLR4, as well as intracellular levels of phospho-p38 MAPK, were unchanged between the two groups (Fig. 6, g–i). Brain-infiltrating T cells showed a reduction in the surface expression of CD69, but the frequency of T cells remained unchanged in the brains of cefepime-treated versus untreated mice (Fig. 6, j and k).

GF mice, which lack any living microorganisms in or on them (Aghighi and Salami, 2024), were used to determine the significance of gut microbiota for CNS-aGVHD development. As littermate controls, we used Oligo-Mouse-Microbiota 12 (OMM¹²)-colonized GF mice, which have a stable microbiota consisting of 12 bacterial strains representing five abundant bacterial phyla in the murine gastrointestinal tract (Brugiroux et al., 2016). Mice underwent lethal irradiation prior to allo-HCT (Fig. 6 l). On D+14, GF and OMM¹² mice displayed similar H2-k^{d+}

donor cell engraftment in the bone marrow (Fig. 6 m). GF mice showed an enlargement of the cecum compared with OMM¹² mice (Fig. 6, n and o). Analysis of the brain revealed an increase in the percentage of CD11b⁺ CD45^{int} cells in GF mice compared with OMM¹² mice (Fig. 6, p and q). The number of Iba-1⁺ myeloid cells was higher in the cortex and cerebellum of GF mice (Fig. 6, r–t). CD11b⁺ CD45^{int} myeloid cells of GF mice exhibited the decreased surface expression of P2RY12 and elevated surface expression of F4/80, indicating their activated phenotype (Fig. 6, u and v). The surface expression of TLR4 and intracellular levels of phospho-p38 MAPK were similar in the two groups (Fig. 6, w and x). The frequency of T cells in the brain and their surface expression of CD69 were also comparable between the groups (Fig. 6, y and z). Taken together, these findings in wildling, GF, and OMM¹² mice demonstrated that microglial activation mediated by antibiotic treatment or gut microbiota depletion is conserved across different gnotobiotic mouse models of aGVHD.

GF mice and antibiotic-treated SPF mice have depleted TMAVA levels

Metabolism and inflammation are two closely linked cellular processes (Gaber et al., 2017). We wanted to assess whether increased microglial activation resulting from microbiota depletion is linked to systemic or microglia-specific changes in metabolites during aGVHD. First, we performed a targeted analysis of polar metabolites on D+14 in the serum, stool, colon, and liver isolated from SPF BALB/c mice that were treated with either antibiotics or vehicle (Fig. 1 a), as well as from GF and OMM¹² mice that underwent allo-HCT as described previously (Fig. 6 l). The bacteria-derived metabolite TMAVA (Liu et al., 2021) was the only metabolite that was decreased in all analyzed tissues in antibiotic- versus vehicle-treated SPF mice (Fig. 7, a–d), as well as in GF versus OMM¹² mice (Fig. 7, e–h). To understand the impact of this systemic TMAVA reduction on microglia, we performed a targeted analysis of polar metabolites in sorted CD11b⁺ CD45^{int} myeloid cells from the brains of antibiotic- or vehicle-treated SPF mice on D+14 after allo-HCT and we discovered that intracellular levels of TMAVA were decreased in

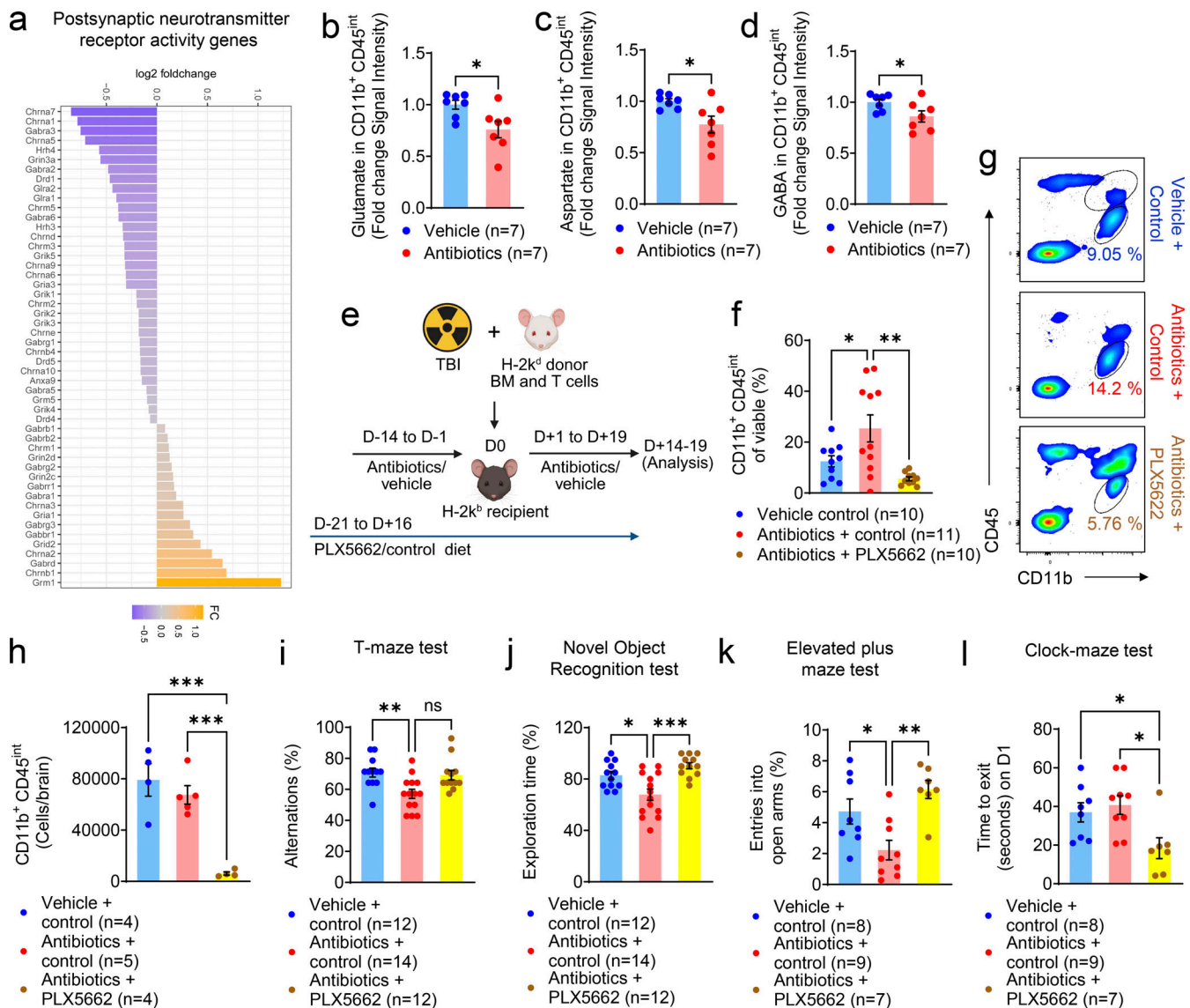


Figure 5. Microbiota depletion impairs cognitive functions after allo-HCT, which can be rescued by pharmacological microglial depletion. (a) Microarray data show top 50 regulated postsynaptic neurotransmitter receptor activity genes in FACS-sorted CD11b⁺ CD45^{int} cells from SPF BALB/c (H2-k^d) recipients treated with antibiotics ($n = 2$) or vehicle ($n = 2$), analyzed on D+14 after allo-HCT. Color codes represent antibiotics versus vehicle log₂ fold change. The experiment was performed once. (b–d) Targeted mass spectrometry analysis of polar metabolites in FACS-sorted CD11b⁺ CD45^{int} cells from SPF BALB/c (H2-k^d) recipients treated with antibiotics or vehicle, on D+14 after allo-HCT. Quantification of intracellular levels of (b) glutamate, (c) aspartate, and (d) GABA. Experiments were performed two times and results (mean $\pm$ SEM) pooled. $n = 7$ per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. * $P < 0.05$. (e–l) Flow cytometry analysis of the brain and cognitive tests on D+14 to D+19 after allo-HCT performed on SPF C57BL/6 (H2-k^b) recipients treated with antibiotics and fed with PLX5622 chow, treated with antibiotics and fed with control chow, or treated with vehicle and fed with control chow. (e) Schematic depiction of treatment and allo-HCT. (f) Quantification of the percentage of CD11b⁺ CD45^{int} cells (f) along with (g) representative flow cytometry plots of each group. (h) Quantification of absolute numbers of CD11b⁺ CD45^{int} cells. Quantification of (i) alternations between left and right arm in the T-maze test, (j) exploration time of the novel object in the novel object recognition test, (k) entries into the open arms of elevated plus maze test, and (l) time taken to exit in the clock maze test. The experiment was performed once (h), two times (f, g, k, and l), or three times (i and j), and results (mean $\pm$ SEM) were pooled. $n = 7$ –14 per group. Each data point represents a biological replicate. P values were calculated using ordinary one-way ANOVA with Tukey's multiple comparisons test or using the Kruskal–Wallis test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, ns = 0.053.

antibiotic- versus vehicle-treated mice (Fig. 7, i and j). Correlation analysis between levels of TMAVA and levels of other detected metabolites in microglia revealed a positive correlation between levels of TMAVA and the electron carrier NADP. We observed a concomitant decrease in NADP levels in sorted CD11b⁺ CD45^{int} cells from the brains of antibiotic-treated mice (Fig. 7, k and l) analyzed by mass spectrometry. Previous

research has shown that TMAVA impacts FAO and mitochondrial function (Zhao et al., 2022), so we wanted to understand whether depletion of TMAVA in microglia was accompanied by changes in metabolic activity and energy production. Microarray analysis on sorted CD11b⁺ CD45^{int} myeloid cells from the brain confirmed a transcriptional upregulation of FAO genes in the antibiotic-treated groups on D+14 (Fig. 7 m). FAO

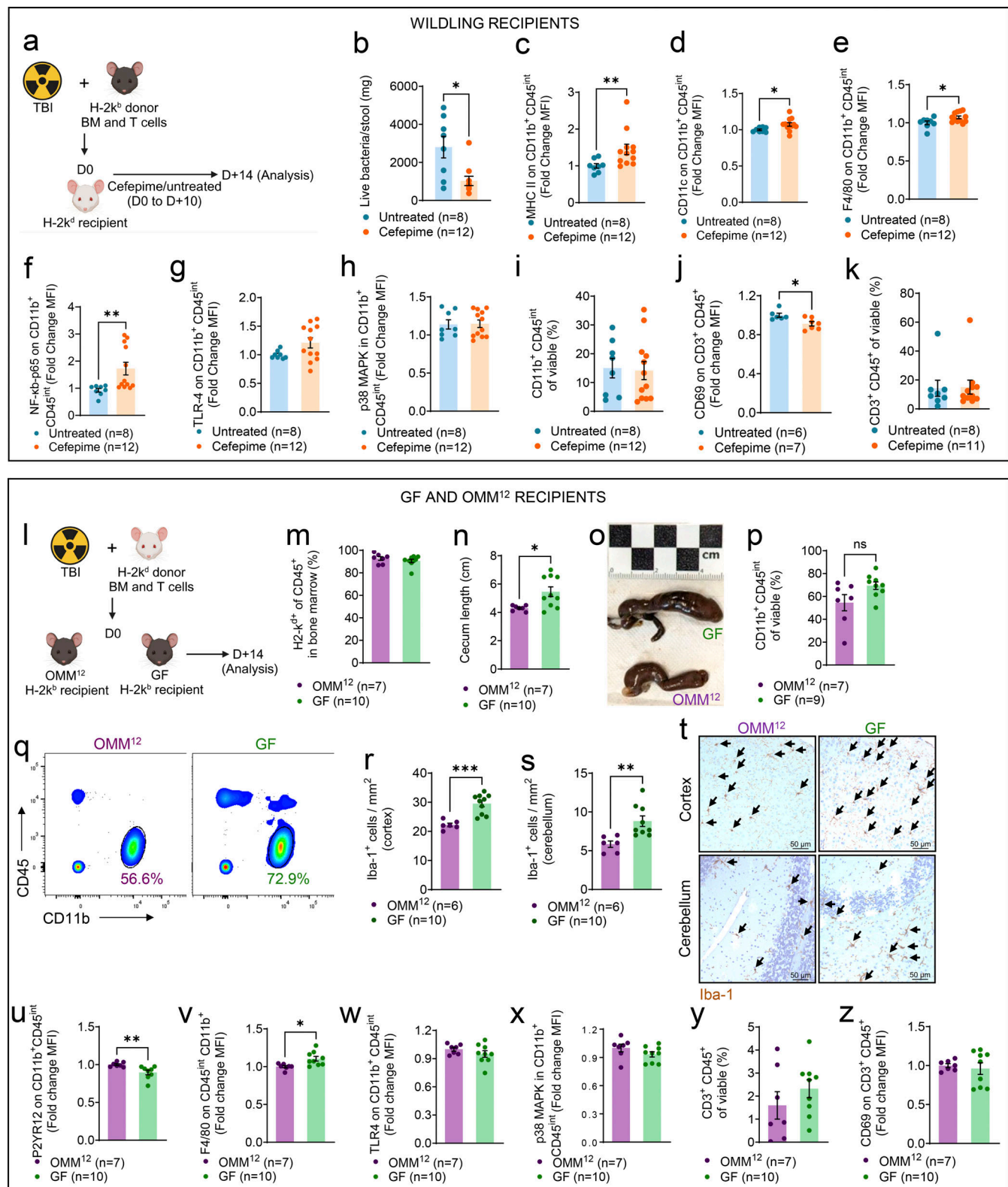


Figure 6. GF versus OMM¹² mice and cefepime versus vehicle-treated wilding mice exhibit microglial activation after allo-HCT. (a and b) Flow cytometry analysis of stool on D+14 from wilding BALB/c (H-2k^d) recipients undergoing allo-HCT and treatment with cefepime or undergoing allo-HCT and no treatment. **(a)** Schematic depiction of treatment and allo-HCT. **(b)** Quantification of the percentage of SYTO 9⁺ bacteria on D+14 (after 10 days of antibiotics/vehicle treatment starting from D0). Experiments were performed two times, and results (mean ± SEM) were pooled. *n* = 8–12 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. **P* < 0.05. **(c–k)** Flow cytometry analysis of the brain on D+14 after allo-HCT from wilding BALB/c (H-2k^d) recipients treated with cefepime or vehicle. Quantification of (c) MHC II, (d) CD11c, (e) F4/80, (f) phospho-NF-κB p65 (Ser536), (g) TLR4, and (h) phospho-p38 MAPK (Thr180, Tyr182) in CD11b⁺ CD45^{int} cells, (i) percentage of CD11b⁺ CD45^{int} cells, (j) CD69 on CD3⁺

CD45 cells, and (k) percentage of CD3⁺ CD45⁺ cells. Experiments were performed two times and results (mean $\pm$ SEM) pooled. $n = 8$ –12 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test or the Mann-Whitney *U* test. * $P < 0.05$, ** $P < 0.01$. **(l–o)** Analysis of bone marrow cells and the cecum on D+14 after allo-HCT from GF and OMM¹² C57BL/6 (H2-k^{b+}) recipients. **(l)** Schematic depiction of treatment and allo-HCT. Quantification of the (m) percentage of H2-k^{d+} cells in the bone marrow using flow cytometry and (n) cecum length along with (o) representative images of each group. The experiment was performed once. $n = 7$ or 10 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. * $P < 0.05$. **(p and q)** Flow cytometry analysis of CD11b⁺ CD45^{int} cells in the brain on D+14 after allo-HCT from GF and OMM¹² C57BL/6 (H2-k^{b+}) recipients. Quantification of the (p) percentage of CD11b⁺ CD45^{int} cells along with (q) representative flow cytometry plots. The experiment was performed once. $n = 7$ or 10 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. ns = 0.06. **(r–t)** Histology of brain samples stained for Iba-1⁺ cells analyzed on D+14 after allo-HCT from GF and OMM¹² C57BL/6 (H2-k^{b+}) recipients. Quantification of absolute number (per mm²) of Iba-1⁺ cells in (r) cerebral cortex and (s) cerebellum, along with (t) representative images of Iba-1⁺ cells (scale bars, 1,000 μ m) in each group. The experiment was performed once. $n = 6$ or 10 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. ** $P < 0.01$, *** $P < 0.001$. **(u–z)** Flow cytometry analysis of the brain on D+14 after allo-HCT from GF and OMM¹² C57BL/6 (H2-k^{b+}) recipients. Quantification of (u) P2RY12, (v) F4/80, (w) TLR4, and (x) phospho-p38 MAPK (Thr180, Tyr182) in CD11b⁺ CD45^{int} cells, (y) percentage of CD3⁺ CD45⁺ cells, and (z) CD69 on CD3⁺ CD45⁺ cells. The experiment was performed once. $n = 7$ or 10 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. * $P < 0.05$, ** $P < 0.01$.

intermediates acetylcarnitine and palmitoylcarnitine were reduced in sorted CD11b⁺ CD45^{int} myeloid cells from antibiotic-treated versus vehicle-treated mice (Fig. S5, a and b). Flow cytometry-based metabolic analysis using the single-cell energetic metabolism by profiling translation inhibition (SCENITH) method (Argüello et al., 2020) on brain cells showed that CD11b⁺ CD45^{int} cells from antibiotic-treated mice had increased FAO and amino acid oxidation (AAO) capacity along with decreased glucose dependence (GD) (Fig. 7, n and o). T cells (Fig. 7, p and q) and CD11b⁺ CD45^{hi} myeloid cells (Fig. S5, c and d) in the brain also exhibited a similar increase in FAO and AAO capacity and decrease in GD.

Since the role of TMAVA in aGVHD has not been described so far, we wanted to understand the impact of the observed TMAVA deficit in CNS-aGVHD and in aGVHD of classical target organs. For this, SPF BALB/c mice were lethally irradiated and underwent allo-HCT and antibiotic treatment as described previously. Additionally, mice were treated with either TMAVA or vehicle via daily intraperitoneal injection (D+1 to D+13) (Fig. 8 a). On D+14, engraftment of H2-k^{b+} donor cells (Fig. 8, b and c) and cecum length (Fig. 8, d and e) was unchanged between TMAVA-treated versus vehicle-treated groups. To assess how TMAVA affects aGVHD severity, we analyzed aGVHD-related mortality (Fig. S5 e), performed clinical scoring of aGVHD symptoms (Fig. S5 f) and histological scoring of aGVHD severity in the colon, liver, and small intestine (Fig. S5, g–i), and found no differences between TMAVA- and vehicle-treated mice. Next, we analyzed microglia and T cells in the brain and observed a decrease in the surface expression of TLR4, as well as a reduction of intracellular phospho-p38 MAPK in CD11b⁺ CD45^{int} myeloid cells of TMAVA-treated mice on D+14 (Fig. 8, f and g). Frequency and numbers of CD11b⁺ CD45^{int} myeloid cells remained unchanged (Fig. S5, j and k), but the frequency of T cells was diminished in the brains of TMAVA-treated mice (Fig. 8, h and i). Numbers of CD3⁺ cells (Fig. 8 j), CD4⁺ T cells, and CD8⁺ T cells (Fig. S5, l and m), intracellular IFN- γ levels in T cells (Fig. S5, n and o), and numbers of IFN- γ ⁺ T cells (Fig. S5, p and q) were similar in the two groups. We also assessed other immune cell subsets in the brain after TMAVA treatment and found a decrease in the frequency of neutrophils (Fig. S5, r and s). The B cell compartment remained unaffected (Fig. S5, t and u). The frequency of CD206⁺ cells decreased in TMAVA-treated mice compared with vehicle-treated mice, but their absolute numbers were unaffected (Fig. S5, v and w).

To elucidate the reason behind decreased TLR4 and phospho-p38 MAPK levels in CD11b⁺ CD45^{int} myeloid cells on TMAVA administration, we first assessed the cause of increased TLR4 expression after microbiota depletion with antibiotic treatment. The absolute number of lipid A⁺ foci in the brain cortex was increased in antibiotic- versus vehicle-treated mice, which was reversed upon TMAVA treatment (Fig. 8, k and l). Concomitantly, we observed increased E-cadherin surface expression on CD31⁺ endothelial cells, which indicated improved tight junction integrity in the BBB (Fig. S5 x) upon TMAVA treatment. To evaluate whether TMAVA reverses metabolic reprogramming of microglia mediated by antibiotic treatment as well, we performed metabolic analysis on brain cells using the SCENITH method after TMAVA treatment and observed a reversal of the increase in FAO and AAO capacity mediated by microbiota depletion (Fig. 8 m).

After we had established an immunomodulatory effect of TMAVA on microglia in vivo in aGVHD mice, we sought to investigate whether TMAVA could also rescue the microglia-mediated cognitive deficits observed in antibiotic-treated aGVHD mice. For this, SPF C57BL/6 mice were treated with antibiotics ad libitum via drinking water (D–14 to D–1), after which they were lethally irradiated and underwent allo-HCT (D0) as described previously. Antibiotic treatment was continued after allo-HCT (D+1 to D+15), and mice were additionally treated with TMAVA or vehicle (D+1 to D+13) (Fig. 8 n). TMAVA-treated mice exhibited an increase in alternations in the T-maze test (Fig. 8 o), increased exploration time of the novel object in the novel object recognition test (Fig. 8 p), and increased time spent in the open arms of elevated plus maze test (Fig. 8 q). These findings demonstrate that microglial TLR4/p38 MAPK pathway activation can be reduced by reversing the systemic TMAVA depletion induced by antibiotic treatment, without affecting overall aGVHD severity. Furthermore, TMAVA administration also alleviated the antibiotic-mediated microglia-dependent cognitive deficits in aGVHD mice.

TMAVA levels are reduced in two independent patient cohorts undergoing allo-HCT, both after the onset of aGVHD and following antibiotic treatment

Previous research has associated the use of certain antibiotics with a higher risk of aGVHD in patients and murine models

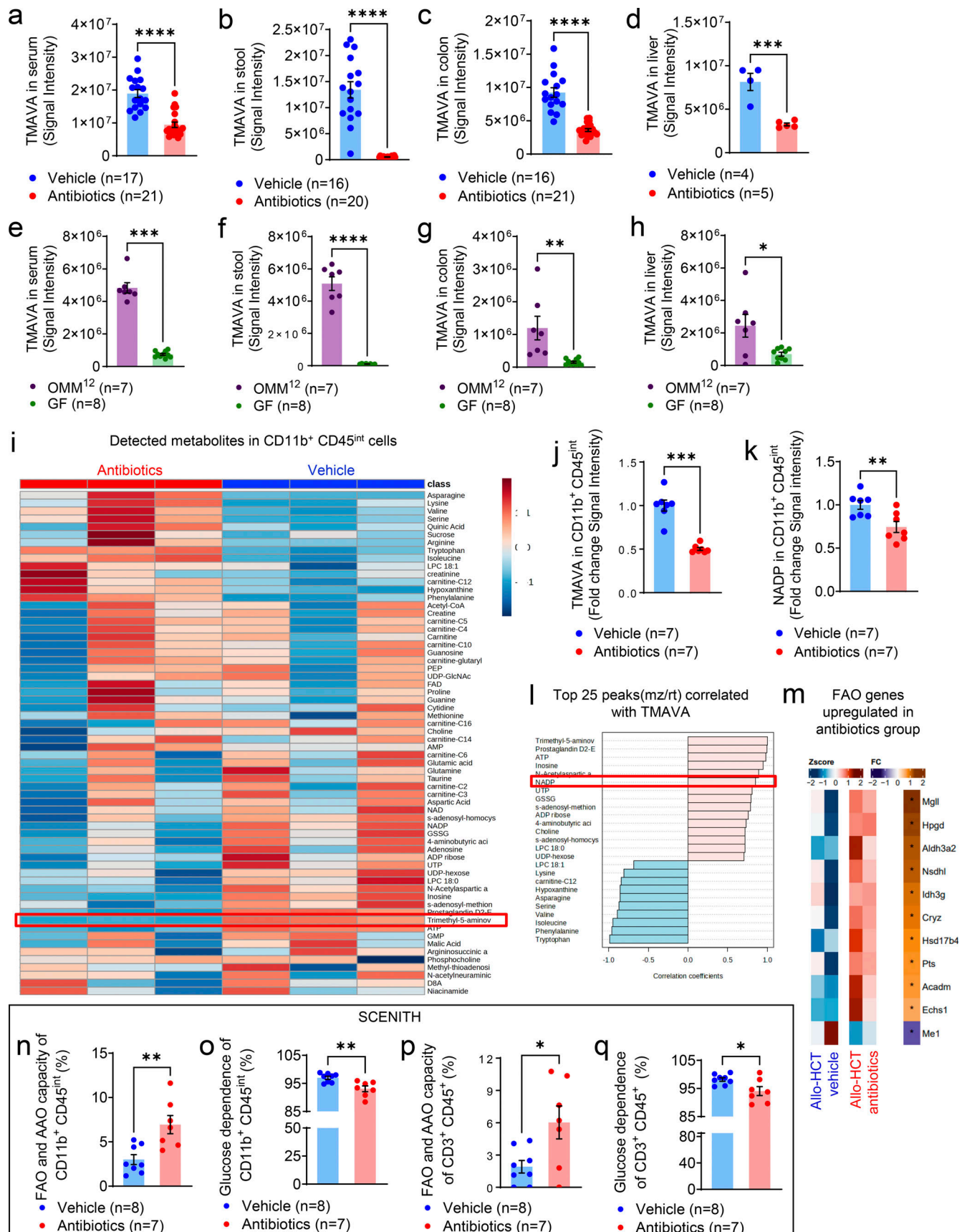


Figure 7. Microglia of microbiota-depleted mice undergo metabolic reprogramming after allo-HCT. (a-d) Targeted mass spectrometry analysis of polar metabolites in different organs from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle, on D+14 after allo-HCT. Quantification of TMAVA in (a)

serum, (b) stool, (c) colon, and (d) liver. The experiment was performed once. The experiment was performed once. $n = 4\text{--}21$ per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired t test or the Mann-Whitney U test. *** $P < 0.001$, **** $P < 0.0001$. **(e–h)** Targeted mass spectrometry analysis of polar metabolites in different organs from GF and OMM¹² C57BL/6 (H2-k^{b+}) recipients, analyzed on D+14 after allo-HCT. Quantification of TMAVA in (e) serum, (f) stool, (g) colon, and (h) liver. The experiment was performed once. The experiment was performed once. $n = 7$ or 8 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired t test or the Mann-Whitney U test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$. **(i–l)** Targeted mass spectrometry analysis of polar metabolites in FACS-sorted CD11b⁺ CD45^{int} cells from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle, on D+14 after allo-HCT. **(i)** Heat map depicts detected metabolites. Quantification of intracellular levels of (j) TMAVA and (k) NADP. **(l)** Correlation analysis shows the top 25 peaks (mz/rt) correlated with TMAVA. Experiments were performed two times and results (mean $\pm$ SEM) pooled. $n = 7$ per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired t test or the Mann-Whitney U test. ** $P < 0.01$, *** $P < 0.001$. **(m)** Microarray analysis shows heat map of differentially regulated FAO genes (*) from FACS-sorted CD11b⁺ CD45^{int} cells from the brains of SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics ($n = 2$) versus vehicle ($n = 2$), analyzed on D+14 after allo-HCT. The experiment was performed once. Color code represents the Z-score log₂ intensity (left panel) and log₂ fold change (right panel). **(n–q)** Flow cytometry-based SCENITH analysis of immune cells in the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle. Quantification of (n) FAO and AAO capacity of CD11b⁺ CD45^{int} cells, (o) GD of CD11b⁺ CD45^{int} cells, (p) FAO and AAO capacity of CD3⁺ CD45⁺ cells, and (q) GD of CD3⁺ CD45⁺ cells. The experiment was performed once. $n = 7$ or 8 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. * $P < 0.05$, ** $P < 0.01$.

(Malard and Jenq, 2023; Shono et al., 2016; Tanaka et al., 2020; Thiele Orberg et al., 2024). To explore the link between antibiotic use and CNS-aGVHD risk, we sought to analyze TMAVA abundance using targeted mass spectrometry in two independent patient cohorts. Cohort 1 consisted of two groups, which underwent allo-HCT and were segregated based on aGVHD occurrence (Michonneau et al., 2019). Cohort 2 consisted of patients analyzed before allo-HCT and after allo-HCT plus antibiotic treatment. Patient characteristics from Cohort 2 are reported in Table 1. We observed a decline in plasma levels of TMAVA in allo-HCT patients developing aGVHD versus those who do not develop aGVHD in Cohort 1 (Fig. 9a). This indicated that GVHD onset after allo-HCT impacts circulatory TMAVA levels in patients. TMAVA was also reduced in the serum of Cohort 2 when compared after allo-HCT and antibiotics to before allo-HCT (Fig. 9, b and c), thus highlighting the impact of antibiotic use on circulatory TMAVA. Pathway enrichment analysis of detected metabolites from Cohort 2 indicated an enrichment of metabolites related to FAO pathways in the serum after allo-HCT and antibiotic treatment (Fig. 9d). These findings demonstrate that both onset of aGVHD and antibiotic use reduce TMAVA levels in the blood of patients undergoing allo-HCT. This aligns with our murine data, which demonstrate reduced TMAVA abundance after antibiotic treatment in SPF mice, as well as in GF mice after allo-HCT.

Discussion

Treatment of CNS-aGVHD is complicated by difficult diagnosis, poor prognosis, and limited therapeutic options (Dowling et al., 2018; Maffini et al., 2017; Vinnakota and Zeiser, 2021). In this study, we investigated whether microbiota depletion exacerbates CNS inflammation and contributes to CNS-aGVHD. We aimed to identify bacterial products and metabolites that modulate CNS-aGVHD and to assess their therapeutic potential in a murine aGVHD model.

Differences in T cell recruitment and local maintenance of immune cell populations determine aGVHD severity in specific organs (Sacicbegovic et al., 2023; Santos et al., 2018). We observed a surge in T cell infiltration in the brains of antibiotic-treated aGVHD mice, demonstrating a regulatory role of the

microbiota in CNS-aGVHD. Previous studies have shown that IFN- γ enhances CD8⁺ T cell migration and priming by APCs (Bhat et al., 2017). Moreover, IFN- γ receptor (IFN- γ R) blockade and subsequent JAK1/JAK2 inhibition can mitigate ongoing aGVHD by reducing antigen presentation and costimulatory molecule expression on recipient APCs (Choi et al., 2018). Studies in mice have also demonstrated that IFN- γ R^{-/-} donor T cells reduce aGVHD-associated mortality (Sun et al., 2012). In line with these findings, we observed an expansion of activated, brain-infiltrating T cells and abundance of CD4⁺ and CD8⁺ T cell-derived IFN- γ in antibiotic-treated mice, which may contribute to exacerbated aGVHD in the CNS.

Microglial activation plays a crucial role in various CNS pathologies, with TLR4 signaling being a key pathway in this process (Rahimifard et al., 2017; Subhramanyam et al., 2019). In neurodegenerative diseases such as Alzheimer's and Parkinson's disease, TLR4 activation on microglia contributes to chronic inflammation and disease progression (Balu et al., 2023; Zhao et al., 2021a; Zhao et al., 2021b; Zhu et al., 2023). Consistent with these findings, we observed increased TLR4 expression on microglia, along with elevated transcriptional activity of the p38 MAPK and NF- κ B pathways in our antibiotic-treated aGVHD mice. We also observed increased LPS levels in the brain cortex in antibiotic-treated mice. Previous studies have shown that bacterial translocation through epithelium can be mediated by changes to epithelial barrier function. Specifically, bacteria such as *Helicobacter pylori* induce E-cadherin ectodomain shedding through host sheddases like ADAM10, leading to the disruption of cell-cell adhesion (Schmidt et al., 2016). Some bacteria, like *Porphyromonas gingivalis*, can downregulate E-cadherin expression at the transcriptional level, further compromising epithelial barrier function (Abe-Yutori et al., 2017). In this study, we observed the decreased expression of E-cadherin on the brain endothelial cells, which could be implicated in increased abundance of LPS in the brain cortex. Increased TLR4 expression on microglia could also be mediated by other TLR4 ligands. It has also been shown that IFN- γ can diminish E-cadherin expression in adherens junctions (Smyth et al., 2012) and in prostate cancer (Korentzelos et al., 2022). The IFN- γ derived from CD4⁺ and CD8⁺ T cells in antibiotic-treated mice could be an additional

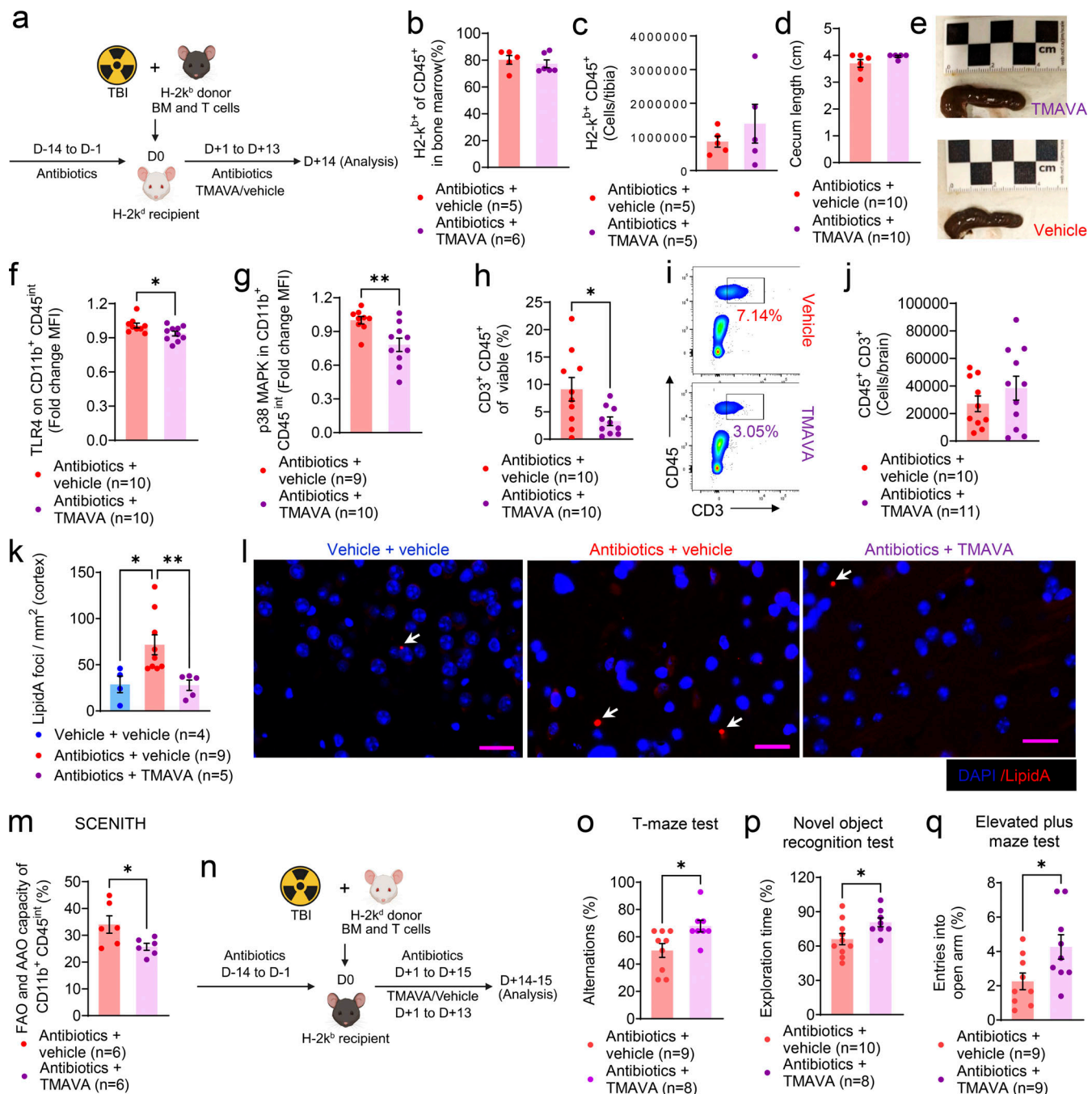


Figure 8. TMAVA rescues microglial metabolic reprogramming after microbiota depletion in allo-HCT. (a–e) Analysis of bone marrow cells and the cecum on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics and TMAVA or vehicle (water). (a) Schematic depiction of treatment and allo-HCT. (b–e) Quantification of (b) percentage and (c) absolute counts of H2-k^{d+} cells in the bone marrow using flow cytometry, and (d) cecum length along with (e) representative images of each group. The experiment was performed once (b and c) or two times (d and e), and results (mean ± SEM) were pooled. The experiment was performed once. *n* = 5–10 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test or the Mann–Whitney *U* test. (f–j) Flow cytometry analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics (all animals) and either TMAVA or vehicle (water). Quantification of (f) TLR4 and (g) phospho-p38 MAPK (Thr180, Tyr182) in CD11b⁺ CD45^{int} cells, (h) percentage of CD3⁺ CD45⁺ cells along with (i) representative flow cytometry plots of indicated groups, and (j) absolute counts of CD3⁺ CD45⁺ cells. Experiments were performed two times (f–j), and results (mean ± SEM) were pooled. The experiment was performed once. *n* = 10 or 11 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test or the Mann–Whitney *U* test. **P* < 0.05, ***P* < 0.01. (k and l) Histology of brain cortex samples stained for lipid A⁺ foci and analyzed on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of the (k) absolute number of lipid A⁺ foci, along with representative images of (l) lipid A⁺ foci (scale bars, 20 μm) in each group. The experiment was performed once. The experiment was performed once. *n* = 4–9 per group. Each data point represents a biological replicate (*n* = 1). P values were calculated using the Kruskal–Wallis test. **P* < 0.05, ***P* < 0.01. (m) Flow cytometry–based SCENITH analysis of CD11b⁺ CD45^{hi} cells in the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics (all animals) and either TMAVA or vehicle (water). Experiments were performed two times, and results (mean ± SEM) were pooled. *n* = 6 per group. Each data point represents a biological replicate (*n* =

1). P values were calculated using two-sided Student's unpaired *t* test. **P* < 0.05. (n–q) Cognitive tests on D+14 and D+15 after allo-HCT performed on SPF C57BL/6 (H2-*k^b*) recipients treated with antibiotics (all animals) and either TMAVA or vehicle (water). (n) Schematic depiction of treatment and allo-HCT. Quantification of (o) alternations between left and right arm in the T-maze test, (p) exploration time of the novel object in the novel object recognition test, and (q) entries into the open arms of elevated plus maze test. Experiments were performed two times and results (mean ± SEM) pooled. *n* = 8–10 per group. Each data point represents a biological replicate. P values were calculated using Student's unpaired *t* test or the Mann–Whitney *U* test. **P* < 0.05.

cause of BBB damage that facilitates increased infiltration of donor-derived immune cells, as well as LPS into the brain.

Neurological complications, which arise in 8–65% of allo-HCT patients, are associated with a poor prognosis and include demyelination, convulsions, motor and sensory symptoms, paralysis, and depression (Maffini et al., 2017; Polchlopek Blasiak et al., 2018; Vinnakota and Zeiser, 2021). In this study, we demonstrated that microbiota depletion exacerbates cognitive deficits in aGVHD mice. We showed that abrogating brain microglia through pharmacological CSF1R inhibition rescued the cognitive deficits caused by antibiotic-mediated microbiota depletion. Studies in various CNS pathologies have shown that microglia modulate cognition and behavior through processes such as neuronal synaptic pruning, ATP gradient generation, and glutamate release (Augusto-Oliveira et al., 2019; Bechade et al., 2013). While this study did not elucidate the specific mechanisms by which microglia directly contribute to cognitive dysfunctions after gut microbiota depletion, we emphasized the critical role of microglia in the development of neurological

aGVHD symptoms in microbiota-depleted aGVHD mice and demonstrated a beneficial effect of microglial depletion on cognitive functions.

SPF mice have a defined microbiota devoid of enteric and respiratory bacterial pathogens such as *Clostridium piliforme*, *Salmonella* spp., and *Mycoplasma pulmonis*, among others (Dobson et al., 2018), resulting in a less complex gut microbiota compared with humans. In contrast, wildling mice harbor a more diverse gut microbiota derived from wild mice, resembling the complexity of the human gut microbiota better than that of conventional SPF mice (Rosshart et al., 2019). On the other hand, GF mice, bred in a sterile environment (Aghighi and Salami, 2024), are entirely devoid of living organisms, allowing researchers to study the effects of its absence on various physiological processes. Previous studies show that GF mice undergoing allo-HCT exhibit higher mortality, weight loss, and clinical GVHD scores than SPF mice (Seike et al., 2023). Although wildling and OMM¹² mice have not been extensively studied for GVHD or CNS pathologies, our findings align with

Table 1. Patient characteristics of Cohort 2 including gender, primary diagnosis, age, antibiotic treatment, and analysis time point

Serial no.	Gender	Primary diagnosis	Age at transplant	Antibiotic treatment	Antibiotics used	Time point of analysis after allo-HCT + antibiotics
1	F	AML NPM1 mutation	72	From D–2	Piperacillin/tazobactam from D–2 to D2 Meropenem from D2	D+4
2	M	AML NPM1 mutation FL3-ITD high	59	From D–2	Piperacillin/tazobactam on D–2 Linezolid from D2	D+4
3	M	MDS-LB	61	From D–4	Piperacillin/tazobactam from D–4 to D1 Meropenem from D1 Vancomycin from D4	D+4
4	F	t-MDS	72	From D0	Piperacillin/tazobactam from D0 to D2 Meropenem from D2	D+8
5	M	AML NPM1 mutation FLT3-TKD	72	From D–2	Piperacillin/tazobactam on D–2	D+6
6	F	Post-PV myelofibrosis	66	From D+2	Piperacillin/tazobactam from D2 to D6 Meropenem from D6	D+11
7	M	MDS-EB2	64	From D–5	Piperacillin/tazobactam from D–5 to D–4 Meropenem from D–4 Tobramycin on D–4 and D–2	D+2

Day 0 (D0) is defined as the day of allo-HCT. F, female; M, male. AML, Acute Myeloid Leukemia; MDS-LB, Myelodysplastic Syndrome with Low Blasts; t-MDS, therapy-related Myelodysplastic Syndrome; PV, Polycythemia Vera.

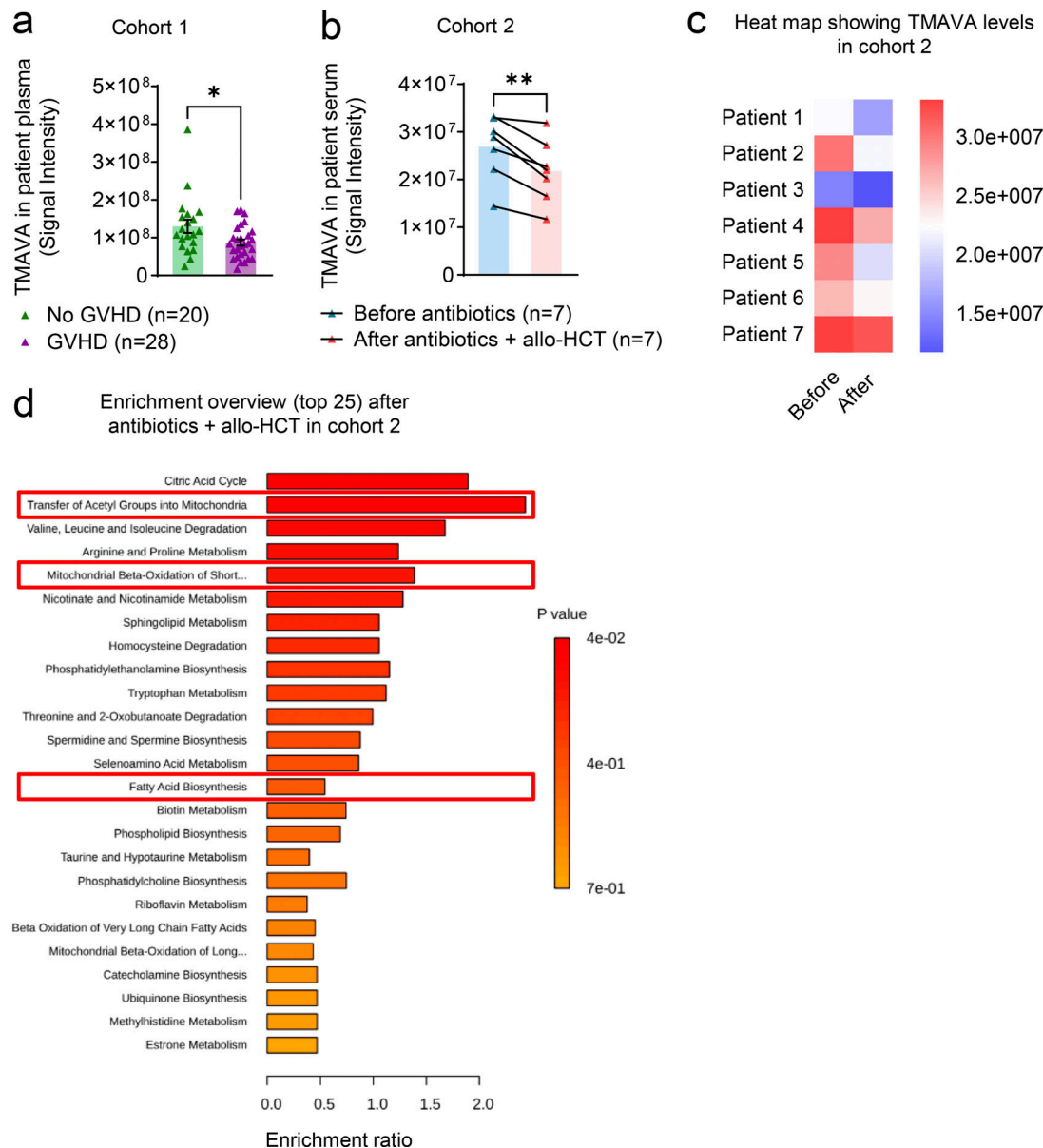


Figure 9. Analysis of patient serum and plasma reveals decline in TMAVA levels after GVHD and after allo-HCT + antibiotics. (a) Targeted mass spectrometry analysis of polar metabolites quantifying TMAVA in plasma from patients without GVHD at D+90 after HCT, and from patients at the onset of GVHD before corticosteroid treatment. The experiment was performed once. $n = 20$ or 28 per group. Each data point represents an individual patient. P values were calculated using the Mann-Whitney U test. $*P < 0.05$. (b-d) Targeted mass spectrometry analysis of polar metabolites in serum from patients at two different time points (before allo-HCT and after allo-HCT and antibiotics for 7–10 days). (b) Quantification of TMAVA in serum. (c) Heat map shows TMAVA in patient sera at two time points. (d) Summary plot for quantitative enrichment analysis of metabolites detected in the serum. The experiment was performed once. Individual patients ($n = 7$) were analyzed at two time points. P values were calculated using a two-sided paired t test. $**P < 0.01$.

previous research that highlighted differences in the structure and function of microglia and other myeloid subsets derived from SPF versus GF or microbiota-depleted mouse brains (Erny et al., 2015; Erny et al., 2021; Sankowski et al., 2021). In this study, we observed increased microglial activation in GF compared with OMM¹² mice, and in cefepime-treated versus untreated wildling mice following allo-HCT, with no significant differences in brain-infiltrating T cells or microglial TLR4 expression. This raises the question of whether microglial TLR4 signaling and T cell infiltration into the brain during aGVHD are influenced by the loss and expansion of certain microbial

keystone communities, which may differ across different gnotobiotic models. Further, the complete absence of bacteria in GF mice differs from the targeted depletion in antibiotic-treated mice. Therefore, the lack of differences in microglial TLR4 and phospho-p38 MAPK levels between GF versus OMM¹² mice and cefepime-treated versus untreated wildling mice may be attributed to less pronounced variations in the microbiota composition between these groups compared with vehicle- versus antibiotic-treated SPF mice. For instance, higher abundance of Clostridiales, Bacteroidetes, and Actinomycetaceae is linked to reduced aGVHD severity and mortality, while expansion of

Enterococcus, *Streptococcus*, Enterobacteriaceae, and *Lactobacillus* is associated with an increased risk of GVHD development and poorer survival (Lin et al., 2021; Yue et al., 2024). Such differences in bacterial species across gnotobiotic models may lead to distinct impacts on host-microbiota interactions and microglial signaling pathways. Although no significant changes in microglial TLR4 or phospho-p38 MAPK levels were observed across the groups, our findings revealed an activated microglial phenotype in GF mice compared with OMM¹² mice, as well as in cefepime-treated versus untreated wildling mice. This underscores that a reduction in the microbiota is sufficient to activate microglia in aGVHD, even though its effect on microglial signaling might vary.

TMAVA is a metabolite produced by gut bacteria (Koistinen et al., 2019; Pessa-Morikawa et al., 2022), and it has not previously been studied in the context of GVHD or allo-HCT. The bacteria *Bifidobacteria* and Coriobacteriaceae, which are linked to TMAVA production, are present in a healthy gut, and their levels are correlated with TMAVA levels in the body (Haikonen et al., 2022). These bacteria also help protect against aGVHD in patients (Beckman et al., 2020; Ebigo et al., 2025). TMAVA has been shown to reduce FAO and acylcarnitine levels both in vitro in human HepG2 cells (Liu et al., 2021), and in vivo in the plasma, liver, and heart of high-fat diet mice (Zhao et al., 2022; Zhao et al., 2020). In line with these findings, in our antibiotic-treated aGVHD mice, we observed a reduction in microglial TLR4/p38 MAPK activity, as well as a reduction in FAO and AAO capacity following TMAVA administration. Moreover, TMAVA reverses the neurocognitive deficits induced by microbiota depletion in aGVHD mice. Our findings indicate that TMAVA reduces TLR4 expression, p38 MAPK pathway activity, and FAO in microglia, which may contribute to the protective effect of TMAVA against neurocognitive deficits caused by the microbiome changes due to antibiotic treatment.

Although we did not observe a systemic effect of TMAVA administration in our murine model, TMAVA may still influence colonic energy metabolism by inhibiting carnitine-dependent FAO in host cells or gut-bacteria, as previously reported (Liu et al., 2021; Zhao et al., 2022; Zhao et al., 2020). This inhibition could shift cellular energy production toward glycolysis, leading to elevation in luminal oxygen levels in the colon. The alleviation of physiological hypoxia may further disrupt the anaerobic niche of obligate anaerobes such as Clostridia, which are known to modulate aGVHD severity (Seike et al., 2023; Simms-Waldrup et al., 2017). Notably, TMAVA has also been implicated in the pathogenesis of heart failure by impairing mitochondrial FAO in cardiomyocytes, thereby contributing to energetic stress and cardiac dysfunction (Zhao et al., 2022). These findings underscore TMAVA's broader potential to interfere with tissue-specific energy metabolism, particularly in organs reliant on FAO for ATP production. Therefore, a therapeutic strategy involving TMAVA should aim to minimize disruption of the gut microbiota, and oral administration should be avoided to prevent alterations in colonic luminal oxygenation.

Previous research has reported increased circulatory levels of TMAVA in patients undergoing fecal microbiota transplant (Liu et al., 2021). Consistent with these findings, we observed a

decrease in TMAVA levels in patient blood after allo-HCT and antibiotic treatment, compared with its levels before allo-HCT. This decrease was accompanied by enrichment of circulating FA metabolites in the blood, reflecting the metabolic shift toward FAO observed in brain-resident myeloid cells of antibiotic-treated aGVHD mice. Plasma levels of TMAVA in aGVHD patients were also lower compared with those in patients without aGVHD, supporting previous studies that have linked gut dysbiosis and loss of anaerobic bacterial species to aGVHD (Lin et al., 2021; Seike et al., 2023).

To conclude, we demonstrated that gut microbiota depletion in aGVHD led to a reduction in TMAVA levels both systemically and in microglia. The reduction in microglial TMAVA was associated with increased activity of the TLR4/p38 MAPK pathway and increased FAO and AAO capacity (Fig. 10). Administering TMAVA to microbiota-depleted aGVHD mice alleviated microglia-mediated neurocognitive deficits, inhibited microglial TLR4/p38 MAPK pathway activity, and diminished FAO and AAO capacity, without affecting systemic aGVHD severity. These findings highlight the immunomodulatory role of TMAVA in CNS-aGVHD, suggesting a potential metabolite-based therapeutic approach.

Materials and methods

Human tissue analysis and patients

Plasma analysis from Cohort 1

Patient details were previously reported (Michonneau et al., 2019).

Serum analysis from Cohort 2

Written consent for clinical research was collected from all patients, and all patient identifiers were removed before analysis. The analysis of human tissue samples was approved by the Ethics Committee of the University of Freiburg (protocol number: 22-1300-S1, Overcoming intestinal immune dysregulation through metabolic modifications during aGVHD). Patient details are shown in Table 1.

Mice

Animal protocols (protocol numbers: G23/100, G22/115, X20/06K) were approved by the Regierungspräsidium Freiburg (regional council), Germany (Federal Ministry for Nature, Environment and Consumers Protection). C57BL/6 (H-2K^b, Thy-1.2) and BALB/c (H-2K^d, Thy-1.2) mice were purchased either from Janvier Labs (France) or from the local stock of the animal facility at the University of Freiburg. C57BL/6 wildling mice (H-2K^b, Thy-1.2) and BALB/c wildling mice (H-2K^d, Thy-1.2) were provided by S.P. Rosshart (Erlangen) and maintained at the animal facility at the University of Freiburg. C57BL/6 (H-2K^b, Thy-1.2) GF and OMM¹² mice and sterile food and water were purchased from S.C. Ganai-Vonarburg (Clean Mouse Facility, University of Bern) and were maintained at the animal facility at the University of Freiburg. All mice were used between 6 and 19 wk of age.

Allo-HCT and GVHD model

SPF recipients: SPF BALB/c (H-2K^d, Thy-1.2) recipient mice were lethally irradiated using a cesium source by two single doses of

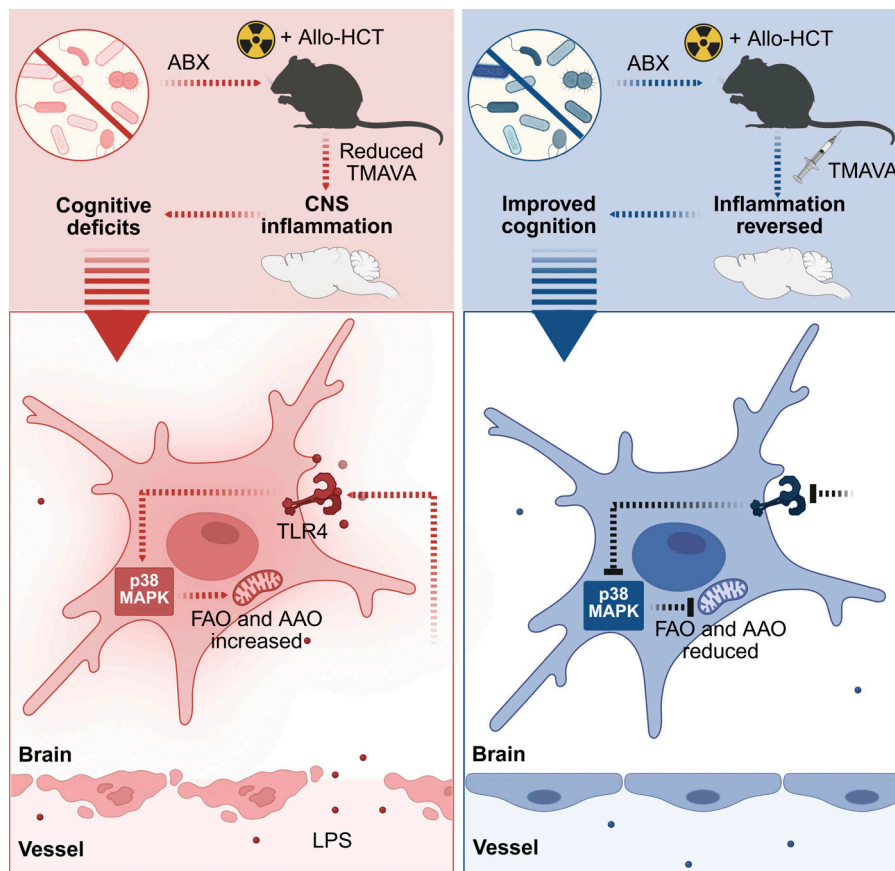


Figure 10. Proposed mechanism of action of TMAVA against murine CNS-aGVHD. Antibiotic treatment leads to systemic and microglia-specific TMAVA reduction and increased LPS infiltration in the brain. Administration of TMAVA reduces microglial TLR4 expression, p38 MAPK pathway activity, and FAO, which contributes to the protective effect of TMAVA against neurocognitive deficits caused by the microbiome changes due to antibiotic treatment.

5 Gy with an interval of at least 4 h. The same day, recipient mice were transplanted with 5×10^6 bone marrow cells and 3×10^5 T cells from MHC-mismatched SPF C57BL/6 (H-2K^b, Thy-1.2) donor mice. SPF BALB/c syngeneic controls were transplanted with 5×10^6 bone marrow cells and 3×10^5 T cells from SPF BALB/c donor mice. For the experiment with allo-HCT using different T cell numbers, mice were injected with either 5×10^6 bone marrow cells only, 5×10^6 bone marrow cells and 1.5×10^5 T cells, 5×10^6 bone marrow cells and 3×10^5 T cells, or 5×10^6 bone marrow cells and 5×10^5 T cells from MHC-mismatched SPF C57BL/6 donor mice. SPF C57BL/6 recipient mice were lethally irradiated using a cesium source by two single doses of 6 Gy with an interval of at least 4 h. The same day, recipient mice were transplanted with 5×10^6 bone marrow cells and 6×10^5 T cells from SPF BALB/c donor mice. SPF C57BL/6 syngeneic controls were transplanted with 5×10^6 bone marrow cells and 6×10^5 T cells from MHC-mismatched SPF C57BL/6 donor mice. Donor T cells from both SPF BALB/c and C57BL/6 mice were purified from the spleens using Pan T Cell Isolation Kit (Miltenyi Biotec). A schematic representation is depicted (Fig. 1 a and Fig. 5 e).

Wildling recipients: Wildling BALB/c (H-2K^d, Thy-1.2) recipient mice were lethally irradiated using an x-ray source by two single doses of 5 Gy with an interval of at least 4 h. The same day, recipient mice were transplanted with 5×10^6 bone marrow cells and 3.5×10^5 T cells from MHC-mismatched wildling C57BL/6 (H-2K^b, Thy-1.2) donor mice. Donor T cells were purified from the spleens using Pan T Cell Isolation

Kit (Miltenyi Biotec). A schematic representation is depicted (Fig. 6 a).

GF and OMM12 recipients: GF and OMM12 C57BL/6 (H-2K^b, Thy-1.2) recipient mice were lethally irradiated using a cesium source by two single doses of 6 Gy with an interval of at least 4 h. The same day, recipient mice were transplanted with 5×10^6 bone marrow cells and 8×10^5 T cells from MHC-mismatched SPF BALB/c (H-2K^d, Thy-1.2) donor mice. Donor T cells were purified from the spleens aseptically in a laminar flow hood using Pan T Cell Isolation Kit (Miltenyi Biotec). A schematic representation is depicted (Fig. 6 l).

Antibiotic treatment

SPF BALB/c recipient mice were treated with antibiotics for 14 days (D-14 to D-1) after which they received allo-HCT on D0 as described previously. The antibiotic treatment was continued for 14 days (D+1 to D+14) after allo-HCT. Antibiotics were administered by daily oral gavage and consisted of 5 mg/ml cefoxitin (Santa Cruz Biotechnology), 5 mg/ml gentamicin (Sigma-Aldrich), 5 mg/ml metronidazole (Sigma-Aldrich), and 5 mg/ml vancomycin (Hexal), dissolved in autoclaved mouse drinking water. Each mouse received 200 μ l of the antibiotics cocktail. As a vehicle, 200 μ l of drinking water was administered by oral gavage. A schematic representation is depicted (Fig. 1 a). On D+14, organs were analyzed by flow cytometry and immunohistochemistry. SPF C57BL/6 recipient mice were also treated with antibiotics for 14 days (D-14 to D-1) after which they received allo-HCT on D0 as described previously. The antibiotic

treatment was continued for 16 days (D+1 to D+16) after allo-HCT. The antibiotics were administered ad libitum via drinking water and consisted of 0.4 mg/ml cefoxitin (Santa Cruz Biotechnology), 0.4 mg/ml gentamicin (Sigma-Aldrich), 0.4 mg/ml metronidazole (Sigma-Aldrich), 0.4 mg/ml vancomycin (Hexal), and 2% wt/vol sucrose dissolved in mouse drinking water. The vehicle mice were given drinking water with 2% weight/volume sucrose (Fig. 5 e). Drinking water was changed every 3–4 days. Behavior studies were performed on D+14, D+15, and D+16 after allo-HCT. For treatment of wildling BALB/c recipient mice with cefepime, an antibiotics stock solution was prepared in 0.9% NaCl (Fresenius Kabi). The mice were injected subcutaneously twice daily with 10 mg/ml cefepime to reach a daily dose of 100 mg/kg body weight from the day of allo-HCT on D0 as described previously, for the next 10 days (D0 to D+10). The control group was untreated (Fig. 6 a). On D+14, organs were analyzed by flow cytometry.

Treatment with CLI-095

SPF BALB/c recipient mice were treated with antibiotics as described previously and with CLI-095 (CAS number: 243984-11-4; InvivoGen) or vehicle for 14 days (D–14 to D–1) after which they received allo-HCT as described previously. The antibiotic treatment, along with CLI-095 or vehicle, was continued for 14 days (D+1 to D+14) after allo-HCT. CLI-095 was injected intraperitoneally at a dosage of 3 mg/kg body weight dissolved in sterile molecular grade water to each mouse daily. Vehicle-treated mice received daily intraperitoneal injections of 3% vol/vol DMSO dissolved in sterile molecular grade water. On D+14, organs were analyzed by flow cytometry and immunohistochemistry. A schematic representation is depicted (Fig. 4 g).

Treatment with TMAVA

SPF BALB/c recipient mice were treated with antibiotics as described previously for 14 days (D–14 to D–1) after which they received allo-HCT on D0 as described previously. The antibiotic treatment was continued for 14 days (D+1 to D+14) after allo-HCT as described previously. Mice were additionally treated with TMAVA (PubChem CID: 14274897, CAS number: 6778-33-2; MedChemExpress) or vehicle for 14 days (D+1 to D+14) after allo-HCT. TMAVA was dissolved to a stock concentration of 1 mg/ml and injected intraperitoneally at a dosage of 3 mg/kg body weight to each mouse daily. Vehicle-treated mice received sterile molecular grade water. On D+14, organs were analyzed by flow cytometry and immunohistochemistry. A schematic representation is depicted (Fig. 8 a).

SPF C57BL/6 recipient mice were treated with antibiotics as described previously for 14 days (D–14 to D–1) after which they receive allo-HCT on D0 as described previously. The antibiotic treatment was continued for 16 days (D+1 to D+16) after allo-HCT as described previously. Mice were additionally treated with TMAVA or vehicle for 14 days (D+1 to D+14) after allo-HCT. TMAVA was dissolved to a stock concentration of 1 mg/ml and injected intraperitoneally at a dosage of 3 mg/kg body weight to each mouse daily. Vehicle-treated mice received sterile molecular grade water. On D+14 and D+15, behavior tests were

performed. On D+16, organs were analyzed by flow cytometry. A schematic representation is depicted (Fig. 8 r).

Microglial depletion

C57BL/6 mice received PLX5622 (CSF1R inhibitor) at 1,200 mg/kg formulated in chow or control chow (CAS number: 1303420-67-8; Research Diets INC.) for 40 days (D–21 to D+19). After 7 days of starting PLX5622 or control chow (D–14), antibiotic treatment ad libitum via drinking water was started as described previously and continued for the next 14 days (D–14 to D–1) after which they receive allo-HCT on D0 as described previously. The antibiotic treatment was continued for 19 days (D+1 to D+19) after allo-HCT as described previously. A schematic representation is depicted (Fig. 5 e).

Immunohistochemistry and immunofluorescence staining

Mice were anesthetized and were perfused with 1× cold PBS. Brains were removed and fixed in 4% phosphate-buffered formalin. Brain tissue was dissected, and sagittal sections were embedded in paraffin. 3-μm sections were prepared and incubated overnight at 37°C before deparaffinization. Epitope retrieval was achieved by citrate buffer (pH = 6) treatment for 40 min. Endogenous peroxidases were blocked with 3% H₂O₂. The sections were then stained with rabbit anti-Iba-1 (1:500, CAS number: 019-19741; Wako) and rat anti-CD3 (1:100, CAS number: MCA500GA; Bio-Rad) antibodies separately for 16 h (1:500) at 4°C for identifying myeloid cells and T cells. The sections were then incubated with biotinylated secondary antibodies anti-rabbit (1:500, CAS number: BA-1000; Vector Laboratories) for Iba-1 and anti-rat (1:200, CAS number: 405-405; BioLegend) for T cells at room temperature for 1 h.

For immunofluorescence labeling of Iba-1⁺ Ki67⁺ cells, 3-μm sagittal FFPE sections were prepared and incubated overnight at 37°C before deparaffinization. Epitope retrieval was achieved by citrate buffer (pH = 6) treatment for 40 min. Nonspecific antibody binding was blocked using blocking solution for goat gold conjugates (CAS number: AU25596; AURION Immuno Gold Reagents & Accessories) at room temperature for 90 min. The sections were then stained with rabbit anti-Ki67 (1:5,000, CAS number: MA5-44693; Invitrogen) and rat anti-Iba-1 (1:500, CAS number: 019-19741; Thermo Fisher Scientific) antibodies at 4°C for 16 h. This was followed by Alexa Fluor 546-conjugated secondary goat anti-rabbit (1:5,000, CAS number: A-11071; Thermo Fisher Scientific) and Alexa Fluor 488 goat anti-rat (1:500, CAS number: A-11071; Invitrogen) antibodies at room temperature for 1 h. Nuclei were counterstained with DAPI (1:5,000) at room temperature for 5 min.

For immunofluorescence labeling of lipid A⁺ foci, 3-μm sagittal FFPE sections were prepared and incubated overnight at 37°C before deparaffinization. Epitope retrieval was achieved by citrate buffer (pH = 6) treatment for 40 min. Nonspecific antibody binding was blocked using 2% BSA in PBS at room temperature for 15 min. TrueBlack(R) Lipofuscin Autofluorescence Quencher (CAS number: 23007; Biomol GmbH) was preheated for 5 min at 70°C and diluted 1:20 in 70% ethanol. Autofluorescence was quenched by incubating sections with diluted quencher and incubated for 10 min at room temperature. The

sections were then stained with goat lipid A LPS antibody (1:50, CAS number: PA1-73178; Thermo Fisher Scientific) at 4°C for 16 h. This was followed by Alexa Fluor 594–conjugated secondary goat anti-rabbit (1:1,000, CAS number: A-11012; Thermo Fisher Scientific) antibody. Nuclei were counterstained with DAPI (1:5,000) at room temperature for 5 min.

Hematoxylin and eosin (Agilent Technologies) staining was performed using a standard protocol on 3- μ m tissue sections.

Microscopy

Images were acquired using ZEISS Axio Imager 2 installed with ZEISS ZEN 2.6 software. For immunohistochemistry staining of CD3⁺ cells, 630 $\times$ magnification was used. For immunofluorescence staining of Iba-1⁺ Ki67⁺ cells and lipid A⁺ foci, 400 $\times$ magnification was used. For immunohistochemistry staining of Iba-1⁺ cells and LFB-PAS, 200 $\times$ magnification was used. Scale bars were added to images using ZEISS ZEN lite software.

Isolation of cells from the murine brain

The CNS cells were isolated as described previously (van Loo et al., 2006) with modifications. Briefly, mice were anesthetized and perfused with 1 $\times$ cold PBS. The whole brain was isolated and homogenized, filtered through a 70- μ m nylon filter, and centrifuged. The homogenate was suspended in 37% isotonic Percoll (GE Healthcare) solution and centrifuged at 800 $\times$ g for 30 min without a break in application. The top myelin layer was removed, and the pelleted cells were washed with 1 $\times$ PBS for further analysis.

For analysis of endothelial cells, meninges were removed by gently rolling the brains over a sterile filter paper. The brains were then homogenized and centrifuged in 37% isotonic Percoll solution before further analysis, as described above.

Isolation of leukocytes from the murine intestine

Intestinal leukocytes were isolated as described previously (Hulsdunker and Zeiser, 2016). Briefly, 4-cm intestinal segments were dissected, and Peyer's patches were removed. The segments were opened longitudinally and rinsed in PBS (Gibco) to remove the remaining feces. Epithelial cells were separated from the lamina propria using cell dissociation buffer (Hanks' balanced salt solution [HBSS; Anprotec] without Mg²⁺, Ca²⁺, 5 mM EDTA [Sigma-Aldrich], 10 mM HEPES [Sigma-Aldrich]). The remaining tissue was digested with digestion buffer (HBSS with Mg²⁺, Ca²⁺, collagenase D 0.5 mg/ml [Roche], DNase 0.5 mg/ml [Qiagen]) to obtain single-cell suspensions, which were then further processed for flow cytometry analyses. Alternatively, tissue was processed by using Mouse Lamina Propria Dissociation Kit (Miltenyi Biotec) and gentleMACS Octo Dissociator with heaters (Miltenyi Biotec) according to the manufacturer's instructions.

T-maze test

For the trial iteration, each mouse was first placed in the starting arm and then allowed to freely choose between the left (L) and right (R) target arm. In the subsequent iteration, the animals were placed back in the starting arm and allowed again to freely choose between the L and R target arm. If the mouse chose the

other arm as compared to the previous iteration, this was counted as a correct alteration. The percentage of these correct arm alterations was calculated from 14 iterations. The 14 test iterations were carried out subsequently, with the maximum iteration duration per animal being 30 min. The number of times the mouse alternated between the two different arms of the maze, in comparison with the total number of possible alternations, was calculated as an index of spatial and working memory. The test was performed once for each mouse.

Novel object recognition test

Each mouse was allowed to explore two similar objects within a total exploration time of 20 s. This was the habituation phase. We started the testing phase 6 h after the habituation phase during which each mouse was allowed to explore the familiar object (to which it was exposed during the habituation phase) and a novel object of different shapes and textures. The position of the novel object and the familiar object was randomized between each mouse. The mouse was allowed to explore the novel object and the familiar object for a total exploration time of 20 s. The amount of time spent by the mouse exploring the novel object in comparison with the total exploration time is calculated as an index of recognition memory. The test was performed once for each mouse (habituation and testing).

Elevated plus maze test

The elevated plus maze test was performed to evaluate the anxiety behavior in mice. Each mouse was kept at the junction of closed arms and open arms of the maze. An open-arm entry was considered when the mouse extended its head toward the open arm or when it entered the arm. The time they spent in the open arm was quantified in minutes. The results were plotted as percentage, showing entry into open arms in relation to the total time. The test was performed once on each mouse.

Clock maze test

The clock maze test was performed to assess spatial memory in mice. On the training day, mice were placed subsequently in a black and then a transparent enclosure with 2 cm deep water, and were trained to find the exit in a clock maze. Mice were trained in each enclosure thrice. On the first test day, the clock maze was filled with 2 cm deep water and time taken to exit was noted. Each mouse was tested four times. The test was continued for 2 more days, with 4 tests per mouse each day (a total of 12 tests over 3 days). The results were plotted as the mean time taken to exit the maze on each day and over 3 days.

Grip strength test

Each mouse was allowed to grasp on to the metal grid of the apparatus, and the tail was pulled backward until they let go of the grid. A total of five trials were performed on each mouse, and the average of grip strength (N) normalized to the body weight of mouse was calculated.

Visual cliff test

The visual cliff test was employed with modifications to evaluate the depth perception of the mice. The test was conducted once

for 10 min per mouse, and each mouse was only tested once. The amount of time spent by the mice in the shallow zone (safe zone) in relation to the total time was calculated.

Flow cytometry

For excluding dead cells, a LIVE/DEAD fixable dead cell stain kit (Molecular Probes) was used according to the manufacturer's instructions. After washing with 1× PBS, cells were incubated with the respective antibodies diluted in FACS buffer for 20 min at 4°C for surface antigen staining after which they were washed and suspended with FACS buffer. 10 µl of flow cytometry cell counting beads (Thermo Fisher Scientific) was added to each sample, and data were acquired on the BD LSRFortessa (BD Biosciences) and analyzed using FlowJo software (Tree Star).

To study murine phospho-p38 MAPK, phospho-NF-κB p65 (CAS number: 115521), and MyD88 (CAS number: sc-136970), cells were first stained with LIVE/DEAD fixable dead cell stain kit followed by surface antigen staining as described previously. Then, they were fixed with 4% formalin for 8 min at 37°C, and subsequently exposed to 90% methanol on ice for 20 min. Cells were then incubated with intracellular antibodies for 1 h at 4°C. Data were acquired on BD LSRFortessa (BD Biosciences) and analyzed using FlowJo software (Tree Star).

For intracellular cytokine staining of murine IFN-γ, BD Cytofix/Cytoperm kit (CAS number: 555028; BD Biosciences) was used. Cells were treated with Cell Stimulation Cocktail (1:500, CAS number: 00-4975; eBioscience) for 4 h prior to staining. Then, cells were first stained with LIVE/DEAD fixable dead cell stain kit followed by surface antigen staining as described previously. After washing with FACS buffer, cells were incubated for 1 h at 4°C with intracellular cytokine antibodies. Data were acquired on the BD LSRFortessa (BD Biosciences) and analyzed using FlowJo software (Tree Star). 10 µl of flow cytometry cell counting beads (CAS number: 01-1234-42; Thermo Fisher Scientific) was added to each sample, and data were acquired on the BD LSRFortessa (BD Biosciences) and analyzed using FlowJo software (Tree Star).

Bacterial load was quantified using flow cytometry as described previously (Mezo et al., 2020), with modifications. Briefly, fecal samples were collected, weighed, homogenized in ice-cold 1× PBS using T 10 basic ULTRA-TURRAX (IKA) homogenizer, and filtered through 40-µm filters. The homogenate was centrifuged at 4,000 rpm for 5 min at 4°C and stained with LIVE/DEAD fixable dead cell stain kit as described previously. Then, cells were incubated with SYTO 9 Green Fluorescent Nucleic Acid Stain (1:1,000 in PBS, CAS number: S34854; Thermo Fisher Scientific) for 15 min at 4°C. 10 µl of flow cytometry cell counting beads (CAS number: 01-1234-42; Thermo Fisher Scientific) was added to each sample, and data were acquired on BD LSRFortessa (BD Biosciences) and analyzed using Flow Jo software (Tree Star).

Microarray data

RNA was isolated from sorted microglial cells using the PicoPure kit (Arcturus). RNA quality was assessed with Agilent 2200 TapeStation (Agilent 141 Technologies). RNA samples with an RNA integrity number >8 were then further processed, and 5 ng

of total RNA was fragmented and converted to labeled ds-cDNA using GeneChip 3' IVT Pico Kit according to the manufacturer's instructions (Applied Biosystems). They were hybridized on Clariom S mouse microarrays (Applied Biosystems) using GeneChip Hybridization, Wash, and Stain Kit (Applied Biosystems). Microarray data were preprocessed using the Robust Multi-Array Average normalization from the oligo R package. Differential gene expression analysis between antibiotics or vehicle (water)-treated H2-*kb*⁺ mice was calculated using the R/Bioconductor limma package (Ritchie et al., 2015) with P values corrected for multiple testing using (Benjamini and Hochberg, 1995). Gene set enrichment analysis was performed using the GAGE R/Bioconductor package (Luo et al., 2009; Castanza et al., 2023) with the MSigDB as gene set resource. The significance threshold was set to 0.05.

Mass spectrometry

Quantification of polar metabolites in sorted CD11b⁺ CD45^{int} cells

After isolation of cells from the murine brain, they were stained for flow cytometry analysis of surface antigens as described elsewhere (Schonberger et al., 2023). Cells were then washed with an aqueous solution containing 0.9% NaCl and subsequently suspended in an aqueous solution containing 0.9% NaCl, 30 µg/ml chlorophenylalanine, and 30 µg/ml aminoterephthalic acid (internal standards). 3,000 CD11b⁺ CD45^{int} cells were sorted directly in 25 µl of extraction buffer containing ¹³C yeast extract as internal standard (ISOTopic solutions) in acetonitrile (LC-MS grade). Fluorescence-activated cell sorting was performed using a 70-µm nozzle and water with 2 g/liter NaCl as sheath fluid. Polar metabolites in sorted cells were quantified using an Agilent 6495 Triple Quadrupole mass spectrometer coupled to an Agilent 1290 Infinity II ultra-high-performance liquid chromatography (UHPLC) system as described elsewhere (Schonberger et al., 2023).

Quantification of polar metabolites in tissues and serum

10 mg of tissue was dissociated with a T 10 basic ULTRA-TURRAX (IKA) homogenizer in 1 ml of methanol: acetonitrile: water solution (50:30:20) prechilled to 4°C. Dissociated tissue was centrifuged at 20,000 × *g* for 10 min at 4°C, and the supernatant was transferred to a DNA LoBind tube (Eppendorf). For quantification of polar metabolites in mouse and human serum, 37 µl of serum was mixed with 500 µl of methanol: water solution (50:50). 250 µl of chloroform was added to this mixture and vortexed briefly. The dissociate was then centrifuged at 20,000 × *g* for 10 min at 4°C, and the clear top phase was transferred to a DNA LoBind tube (Eppendorf). Polar metabolites were quantified using an Agilent 6495 Triple Quadrupole mass spectrometer coupled to an Agilent 1290 Infinity II UHPLC system.

Quantification of SCFAs in tissues and serum

50 mg of tissue was dissociated with a T 10 basic ULTRA-TURRAX (IKA) homogenizer in 1 ml of acetonitrile: water solution (50:50) prechilled to 4°C. The dissociated tissue was centrifuged at 20,000 × *g* for 10 min at 4°C, and the supernatant was transferred to a DNA LoBind tube (Eppendorf). For quantification of polar metabolites in serum, 50 µl of serum was mixed with 50 µl of acetonitrile solution and vortexed briefly. The

dissociate was then centrifuged at $20,000 \times g$ for 10 min at 4°C , and the supernatant was transferred to a DNA LoBind tube (Eppendorf). SCFAs were quantified using an Agilent 6495 Triple Quadrupole mass spectrometer coupled to an Agilent 1290 Infinity II UHPLC system.

For all mass spectrometry experiments, raw area/height of chromatographic peaks (signal intensity) was used as a measure of metabolite abundance.

Assessing metabolic fitness of microglia ex vivo using SCENITH

Cells were isolated from the murine brain as described previously and grouped into four conditions with 2.5×10^5 cells in each, which were subsequently processed for the SCENITH assay as described elsewhere (Argüello et al., 2020; Suhail et al., 2023) with modifications. Briefly, the four treatment conditions were (1) control, (2) 2-deoxy-D-glucose (2DG; final concentration 100 mM) treated, (3) oligomycin (OM; final concentration 1 mM) treated, and (4) 2DGOM (2DG, first and then OM, combined) treated for 30 min at room temperature. Each treatment group then received OPP (10 $\mu\text{g}/\text{ml}$; InvivoGen) during the final 15 min of the treatment. After washing with cold PBS, cells were stained for flow cytometry analysis of surface antigens CD11b, CD45, and CD3 as described previously. After washing of antibodies, cells were fixed and permeabilized using fixation/permeabilization buffer set (BD Biosciences) as described previously. Intracellular staining of OPP was performed by incubating with anti-Puro AF647 antibody created using Click-iT Protein Reaction Buffer Kit (1:100, CAS number: C10458; Thermo Fisher Scientific) for 30 min at room temperature in the dark. Cells were analyzed using flow cytometry for MFI of incorporated OPP.

Statistics and reproducibility

Data were tested for outliers using the ROUT method ($Q = 1\%$), and outliers were excluded from analysis. Additionally, data were tested for normality using the Shapiro-Wilk, D'Agostino-Pearson, and Anderson-Darling tests. For normally distributed data, an unpaired two-sided Student *t* test was applied for two groups. If the data were not distributed normally, the Mann-Whitney *U* test was applied for two groups. If >2 groups were analyzed, ordinary one-way ANOVA was used for normally distributed data and the Kruskal-Wallis test was used in case of data that are not distributed normally. Correlation analysis was performed using simple linear regression. Survival analysis was performed using the Mantel-Cox test. Statistical analyses were done using GraphPad Prism 10.0 (GraphPad Software). For metabolomics datasets, heat maps were created using MetaboAnalyst 6.0 and GraphPad Prism 10.0. Correlation analysis using pattern searching and pathway analysis using a defined background set were performed using MetaboAnalyst 6.0. Data were presented as the mean and SEM (error bars). $P < 0.05$ was reported as statistically significant. The experiments were performed in a nonblinded fashion except for the analysis of human and mouse tissue specimens and behavior tests.

Online supplemental material

Supplemental figures include data on CNS-infiltrating immune cells in antibiotic- versus vehicle-treated aGVHD mice and

immune cells in the blood and spleen of antibiotic- versus vehicle-treated aGVHD mice, and the effect of PLX5622 treatment on different immune cell subsets in the bone marrow, spleen, and brain isolated from mice treated with antibiotics and fed with PLX5622 chow or mice treated with antibiotics and fed with control chow. Fig. S1 shows flow cytometry plots for engraftment, infiltrating immune cells, and endothelial cells. Fig. S2 shows flow cytometry plots for T cells in the blood and spleen, Iba-1⁺ cell counts in the brain, and flow cytometry plots for microglia. Fig. S3 shows flow cytometry plots for colon lamina propria and epithelium. Fig. S4 shows LFB-PAS staining in the brain, and flow cytometry plots for spleen and bone marrow on PLX5622 treatment. Fig. S5 shows mass spectrometry analysis of microglia and SCENITH assay on myeloid cells in the brain.

Data availability

Microarray data have been uploaded on Gene Expression Omnibus and are accessible under the id: GSE279739. Mass spectrometry data have been uploaded on MassIVE (<https://massive.ucsd.edu/ProteoSAFe/static/massive.jsp>) under accession no. MSV000096417. All other data are available from the authors upon request.

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Supplemental material

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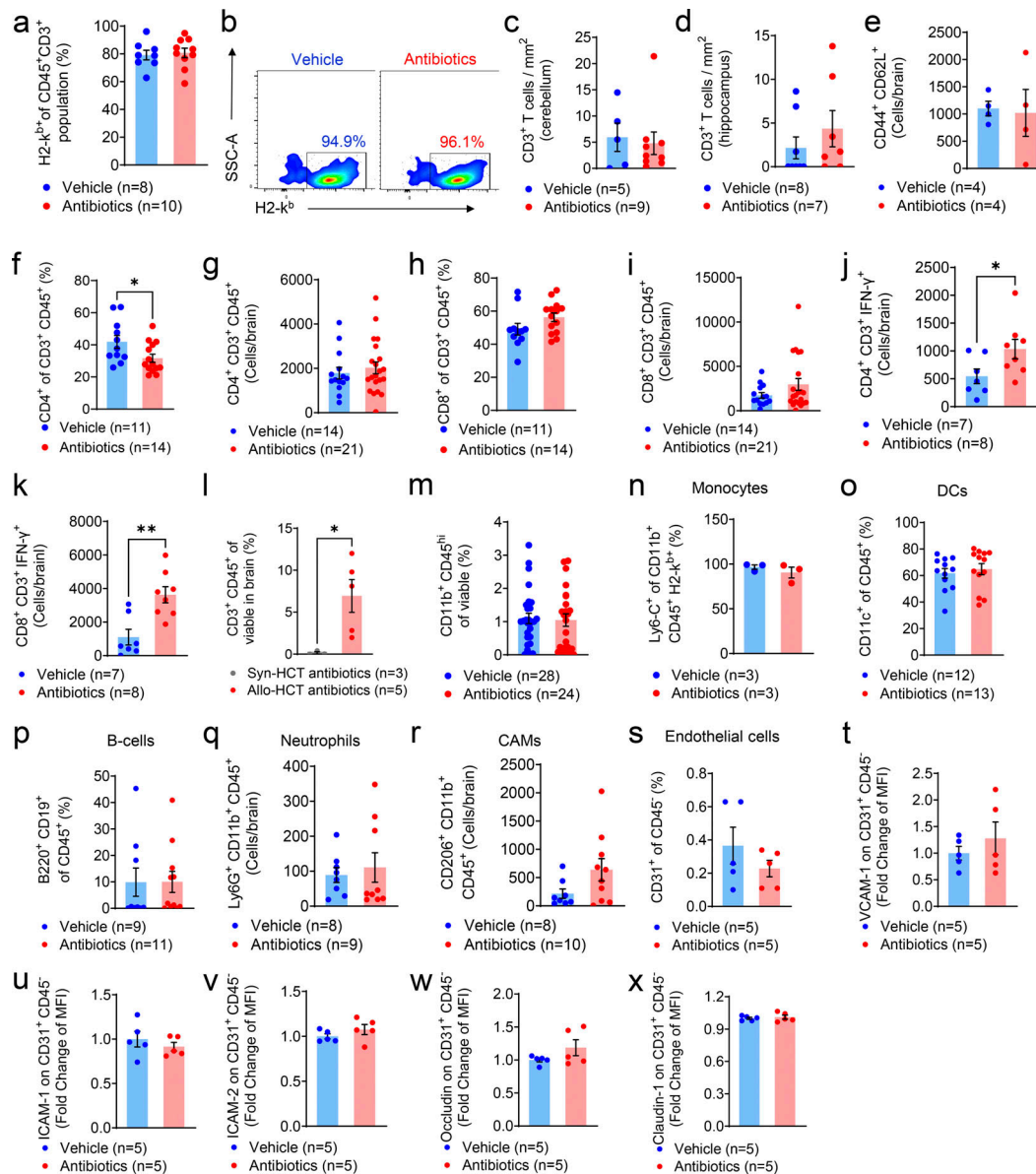


Figure S1. Microbiota depletion does not affect donor cell engraftment in the bone marrow or T cell subpopulations infiltrating the brain. (a and b) Flow cytometry analysis of bone marrow cells on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). **(a and b)** Quantification of the percentage of H2-k^{d+} cells in the bone marrow (a) along with (b) representative flow cytometry plots of each group. Experiments were performed two times and results (mean ± SEM) pooled. *n* = 8 or 10 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. **(c and d)** Histology of brain samples stained for CD3⁺ cells analyzed on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of the absolute number (per mm²) of CD3⁺ cells in the (c) cerebellum and (d) hippocampus. Experiments were performed two times or three times. *n* = 5–9 per group. Each data point represents a biological replicate. P values were calculated using the Mann–Whitney *U* test. **(e–k)** Flow cytometry analysis of different CD3⁺ cell subsets in the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of (e) absolute counts of CD44⁺ CD62L⁺ cells, (f) percentage of CD3⁺ CD4⁺ cells, and (h) CD3⁺ CD8⁺ cells, absolute counts of (g) CD3⁺ CD4⁺ cells, (i) CD3⁺ CD8⁺ cells, (j) CD3⁺ CD4⁺ IFN-γ⁺ cells, and (k) CD3⁺ CD4⁺ IFN-γ⁺ cells. The experiment was performed once (e), two times (j and k), three times (f and h), or five times (g and i) and results (mean ± SEM) pooled. *n* = 4–21 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test or the Mann–Whitney *U* test. **P* < 0.05, ***P* < 0.01. **(l)** Flow cytometry analysis of the brain on D+14 after allo-HCT or syn-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics. Quantification of the percentage of CD3⁺ CD45⁺ cells. The experiment was performed once. *n* = 3 or 5 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. **(m–r)** Flow cytometry analysis of different CD45⁺ cell subsets in the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of the percentage of (m) CD11b⁺ CD45^{hi} cells, (n) Ly6C⁺ cells, (o) CD11c⁺ cells, (p) B220⁺ CD19⁺ cells, and absolute counts of (q) Ly6G⁺ cells and (r) CD206⁺ cells. The experiment was performed once (n), two times (p–r), three times (o), or nine times (m) and results (mean ± SEM) pooled. *n* = 3–28 per group. Each data point represents a biological replicate. P values were calculated using either two-sided Student's unpaired *t* test or the Mann–Whitney *U* test. **(s–x)** Flow cytometry analysis of CD31⁺ CD45[−] cells in the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of the (s) percentage of CD31⁺ CD45[−] cells, (t) VCAM-1, (u) ICAM-1, (v) ICAM-2, (w) occludin, and (x) claudin-1 on CD31⁺ CD45[−] cells. The experiment was performed once. *n* = 5 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test.

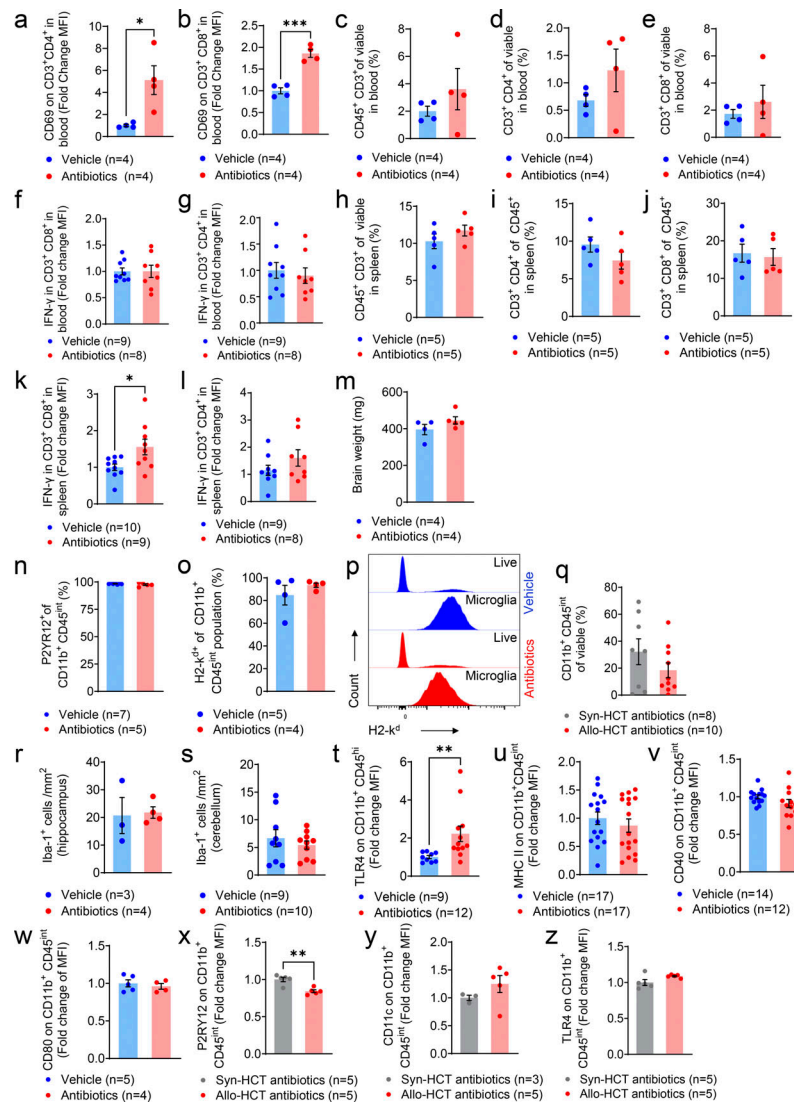


Figure S2. Microbiota depletion does not affect circulating and splenic T cell populations, as well as costimulatory molecule expression on microglia in aGVHD. (a–l) Flow cytometry analysis of the spleen and PBMCs isolated from the blood on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of (a) CD69 on CD3⁺ CD4⁺ cells, (b) CD69 on CD3⁺ CD8⁺ cells, (c) percentage of CD3⁺ CD45⁺ cells, (d) percentage of CD3⁺ CD4⁺ cells, (e) percentage of CD3⁺ CD8⁺ cells, (f) IFN- γ in CD3⁺ CD8⁺ cells, and (g) IFN- γ in CD3⁺ CD4⁺ cells in PBMCs isolated from the blood. Quantification of the (h) percentage of CD3⁺ CD45⁺ cells, (i) percentage of CD3⁺ CD4⁺ cells, (j) percentage of CD3⁺ CD8⁺ cells, (k) IFN- γ in CD3⁺ CD8⁺, and (l) IFN- γ in CD3⁺ CD4⁺ cells in the spleen. The experiment was performed once (a–e and h–j) or two times (f, g, k, and l). $n = 4–10$ per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. * $P < 0.05$, *** $P < 0.001$. **(m)** Quantification of the brain weight on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). The experiment was performed once. $n = 4$ per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. **(n–p)** Flow cytometry analysis of CD45^{int} CD11b⁺ cells in the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of percentage of (n) P2RY12 and (o) H2-k^{d+} cells, along with (p) representative histograms of each group. The experiment was performed once (o) or two times (n) and results (mean $\pm$ SEM) pooled. $n = 4–7$ per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. **(q)** Flow cytometry analysis of CD11b⁺ CD45^{int} cells in the brain on D+14 after allo-HCT or syn-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics. Quantification of the percentage of CD45^{int} CD11b⁺ cells. Experiments were performed two times and results (mean $\pm$ SEM) pooled. $n = 8–10$ per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. **(r and s)** Quantification of brain samples stained for Iba-1⁺ cells and analyzed on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of the absolute number (per mm²) of Iba-1⁺ cells in the (r) hippocampus and (s) cerebellum. The experiment was performed once (r) or three times (s) and results (mean $\pm$ SEM) pooled. $n = 3–10$ per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. **(t–w)** Flow cytometry analysis of CD11b⁺ CD45^{hi} cells and CD11b⁺ CD45^{int} cells in the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of (t) TLR4 on CD11b⁺ CD45^{hi} cells, (u) MHC II on CD11b⁺ CD45^{int} cells, (v) CD40 on CD11b⁺ CD45^{int} cells, and (w) CD80 on CD11b⁺ CD45^{int} cells. The experiment was performed once (w), or four times (t and v), or four times (u) and results (mean $\pm$ SEM) pooled. $n = 4–17$ per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test or the Mann-Whitney U test. ** $P < 0.01$. **(x–z)** Flow cytometry analysis of CD11b⁺ CD45^{int} cells in the brain on D+14 after allo-HCT or syn-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics. Quantification of (x) P2RY12, (y) CD11c, and (z) TLR4. The experiment was performed once. $n = 3$ or 5 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. ** $P < 0.01$.

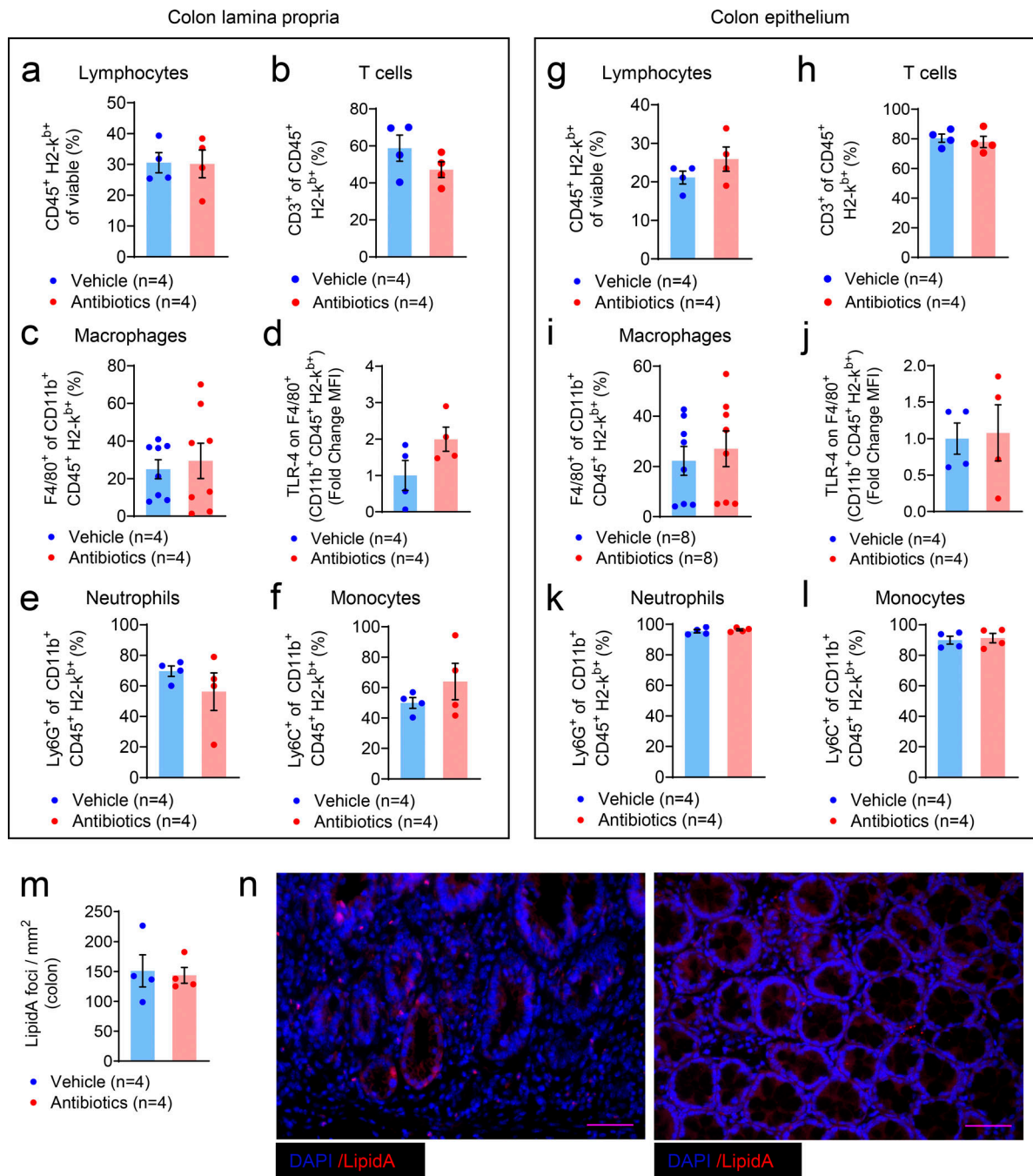


Figure S3. Microbiota depletion does not affect myeloid and lymphoid compartments in the colon lamina propria or epithelium. (a–f) Flow cytometry analysis of H2-k^{b+} cells in the colon lamina propria on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of (a) percentage of CD45⁺ cells, (b) percentage of CD3⁺ CD45⁺ cells, (c) percentage of F4/80⁺ cells, (d) TLR4 on F4/80⁺ cells, (e) percentage of Ly6G⁺ cells, and (f) percentage of Ly6C⁺ cells. The experiment was performed once. *n* = 4 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. **(g–l)** Flow cytometry analysis of H2-k^{b+} cells in the colon epithelium on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of (g) percentage of CD45⁺, (h) percentage of CD3⁺ CD45⁺ cells, (i) percentage of F4/80⁺ cells, (j) TLR4 on F4/80⁺ cells, (k) percentage of Ly6G⁺ cells, and (l) percentage of Ly6C⁺ cells. The experiment was performed once. *n* = 4 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test. **(m and n)** Histology of colon samples stained for lipid A⁺ foci and analyzed on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle (water). Quantification of (m) absolute number of lipid A⁺ foci along with (n) representative images (scale bars, 50 μ m) in each group. The experiment was performed once. *n* = 4 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired *t* test.

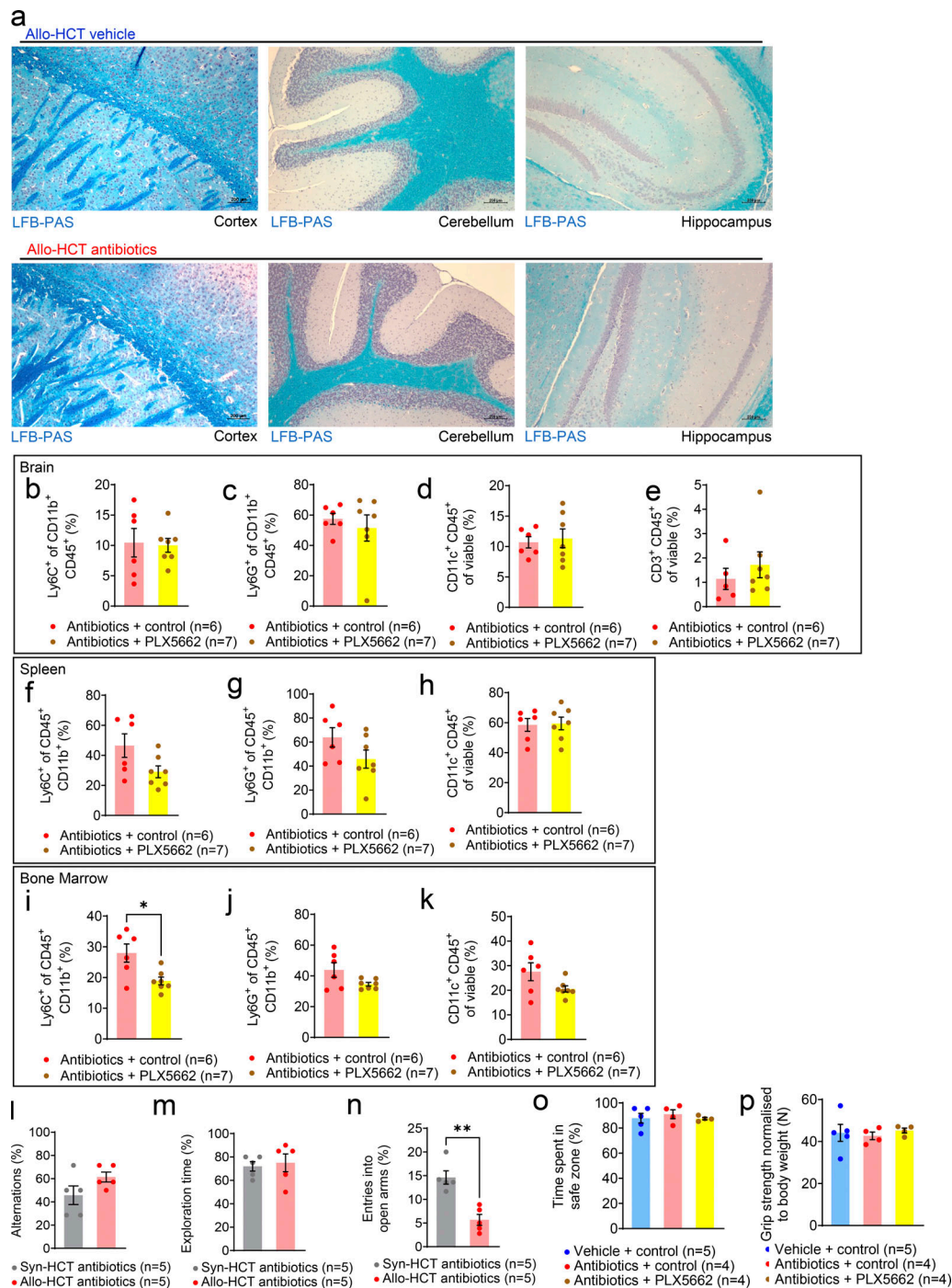


Figure S4. Microbiota depletion does not affect myelination in the brain, and pharmacological CSF1R inhibition does not alter immune cell populations in either the brain or periphery. (a) Representative images of each group for histology of brain samples stained for LFB-PAS on D+14 after allo-HCT from SPF BALB/c (H2-K^d) recipients treated with antibiotics or vehicle (water) (scale bars, 200 μ m). $n = 4$ analyzed per group. (b–k) Flow cytometry analysis of different organs on D+16 after allo-HCT performed on SPF C57BL/6 (H2-K^b) recipients treated with antibiotics and fed with PLX5622 or control chow. Quantification of percentage of (b) Ly6C⁺, (c) Ly6G⁺, (d) CD11c⁺, and (e) CD3⁺ CD45⁺ cells in the brain. Quantification of percentage of (f) Ly6C⁺, (g) Ly6G⁺, and (h) CD11c⁺ in the spleen. Quantification of percentage of (i) Ly6C⁺, (j) Ly6G⁺, and (k) CD11c⁺ cells in the bone marrow. The experiment was performed once. $n = 6$ or 7 per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. * $P < 0.05$. (l–n) Cognitive tests on D+16 after allo-HCT or syn-HCT performed on SPF C57BL/6 (H2-K^b) recipients treated with antibiotics. Quantification of (l) alternations between left and right arm of a T-maze test, (m) exploration time of the open arms in elevated plus maze test, and (n) entries into the open arms of elevated plus maze test. The experiment was performed once. $n = 5$ per group. Each data point represents a biological replicate. P values were calculated using two-sided Student's unpaired t test. ** $P < 0.01$. (o and p) Cognitive tests on D+14 to D+19 after allo-HCT performed on SPF C57BL/6 (H2-K^b) recipients treated with antibiotics and fed with PLX5622 chow, treated with antibiotics and fed with control chow, or treated with vehicle and fed with control chow. (o) Time spent in open zone of the visual cliff test and (p) grip strength. The experiment was performed once. $n = 4$ or 5 per group. Each data point represents a biological replicate. P values were calculated using ordinary one-way ANOVA with Tukey's multiple comparisons test.

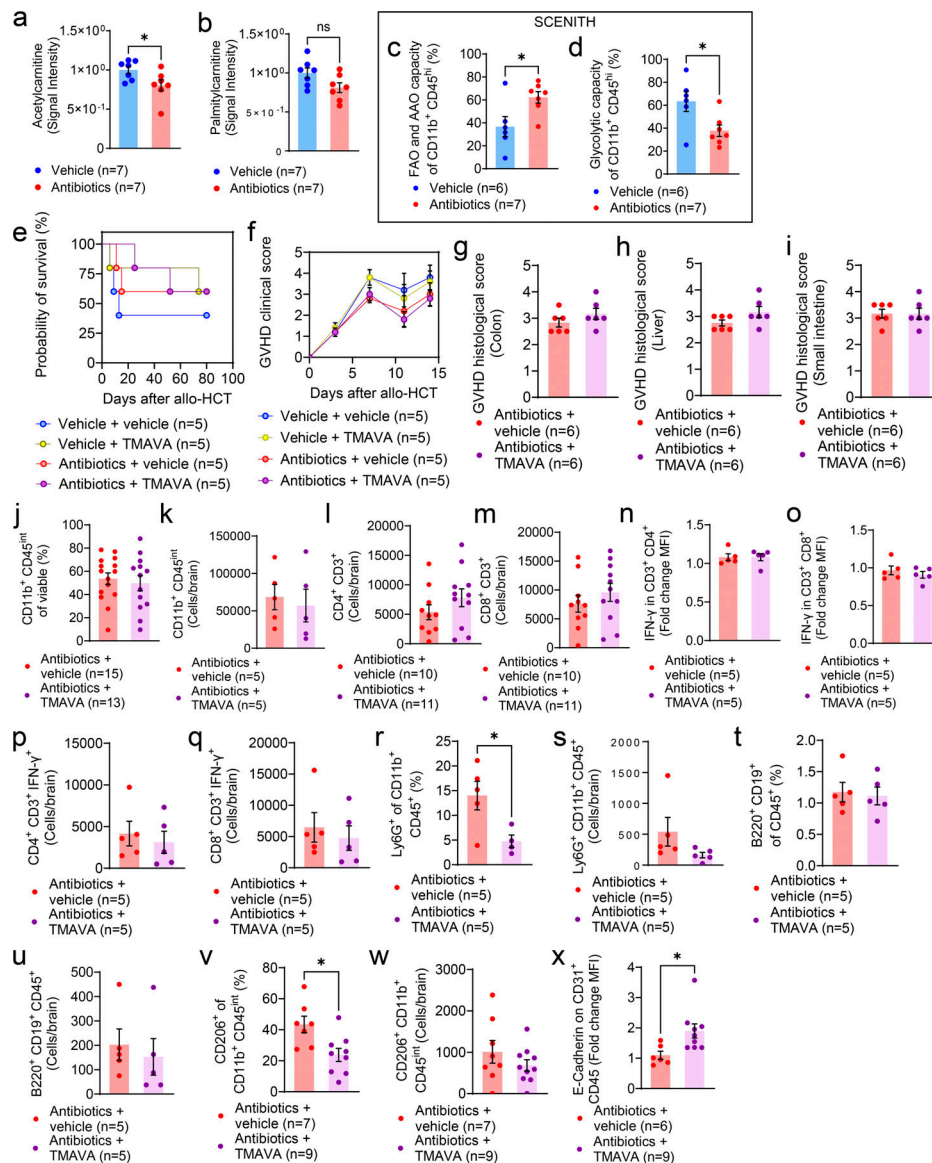


Figure S5. Microbiota depletion affects metabolic activity of CD11b⁺ CD45^{hi} myeloid cells in the brain, and TMAVA administration does not affect T cell populations in the periphery. (a and b) Targeted mass spectrometry analysis of polar metabolites in FACS-sorted CD11b⁺ CD45^{int} cells from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle, on D+14 after allo-HCT. Quantification of intracellular levels of (a) acetylcarnitine and (b) palmitoylcarnitine. Experiments were performed two times and results (mean ± SEM) pooled. *n* = 7 per group. Each data point represents a biological replicate. *P* values were calculated using either two-sided Student's unpaired *t* test. **P* < 0.05, ns = 0.059. (c and d) Flow cytometry-based SCENITH analysis of CD11b⁺ CD45^{hi} cells in the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics or vehicle. Quantification of (c) FAO and AAO capacity of CD45^{hi} CD11b⁺ cells, and (d) GD of CD11b⁺ CD45^{hi} cells. The experiment was performed once. *n* = 6 or 7 per group. Each data point represents a biological replicate. *P* values were calculated using two-sided Student's unpaired *t* test. **P* < 0.05. (e) GVHD-related mortality of SPF BALB/c (H2-k^{d+}) recipients undergoing allo-HCT and treated with antibiotics (all animals) and either TMAVA or vehicle (water). The experiment was performed once. *n* = 5 per group. Each data point represents a biological replicate. *P* values were calculated using the Mantel-Cox test. The Bonferroni-corrected threshold was set as 0.017. (f) Clinical scoring of aGVHD symptoms in SPF BALB/c (H2-k^{d+}) recipients undergoing allo-HCT and treated with antibiotics (all animals) and either TMAVA or vehicle (water). The experiment was performed once. *n* = 5 per group. Each data point represents a biological replicate. *P* values were calculated using ordinary one-way ANOVA with Tukey's multiple comparisons test. Bonferroni-corrected threshold was set as 0.017. (g-i) Histological scoring of GVHD severity in colon, small intestine, and liver samples stained for H&E on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics (all animals) and either TMAVA or vehicle (water). Quantification of GVHD histological scores in the (g) colon, (h) liver, and (i) small intestine. The experiment was performed once. *n* = 6 per group. Each data point represents a biological replicate. *P* values were calculated using two-sided Student's unpaired *t* test. (j-x) Flow cytometry analysis of the brain on D+14 after allo-HCT from SPF BALB/c (H2-k^{d+}) recipients treated with antibiotics (all animals) and either TMAVA or vehicle (water). Quantification of (j) percentage of CD11b⁺ CD45^{int} cells, and absolute counts of (k) CD11b⁺ CD45^{int} cells, (l) CD3⁺ CD4⁺ cells and (m) CD3⁺ CD8⁺ cells. Quantification of IFN-γ in (n) CD3⁺ CD4⁺ cells and (o) CD3⁺ CD8⁺ cells. Absolute counts of (p) CD3⁺ CD4⁺ IFN-γ⁺ cells, and (q) CD3⁺ CD8⁺ IFN-γ⁺ cells, (r) percentage of Ly6G⁺ cells, (s) absolute counts of Ly6G⁺ cells, (t) percentage of B220⁺ CD19⁺ cells, (u) absolute counts of CD19⁺ B220⁺ cells, (v) percentage of CD206⁺ cells, (w) absolute counts of CD206⁺ cells, and (x) E-cadherin on CD31⁺ CD45⁻ cells. The experiment was performed once (k and n-t), two times (l, m, and u-x), or three times (j). *n* = 5–15 per group. Each data point represents a biological replicate. *P* values were calculated using two-sided Student's unpaired *t* test.